`bispectrum`: Selective G-Bispectra Made Practical

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Abstract

Many machine-learning tasks are invariant under group actions: image classification under planar rotation $\mathrm{SO}(2)$, signal classification under translation, and spherical-image classification under 3D rotations $\mathrm{SO}(3)$. The G-bispectrum is a principled *complete* invariant (it preserves all signal information up to the group action) but its deployment has been hampered by quadratic cost and a patchwork of group-specific implementations. We present `bispectrum`, an open-source, fully unit-tested PyTorch library that computes *selective* G-bispectral invariants for seven different group/domain pairs as drop-in differentiable `nn.Modules`. Selectivity reduces the invariant size from $O(|G|^2)$ for the full bispectrum to $O(|G|)$ while remaining complete on generic signals. On the theoretical side, we contribute a selective $\mathrm{SO}(3)$ invariant on the sphere S^2 that achieves $\Theta(L^2)$ coefficients at band-limit L, matching the information-theoretic orbit-space lower bound $(L{+}1)^2 - 3$ instead of the $O(L^3)$ of the full bispectrum. We give a Kakarala-style conditional completeness theorem reducing the question to two cleanly named hypotheses (finite-band seed injectivity and uniform bootstrap-determinant nonvanishing), both supported by per-degree numerical witnesses up to $\ell = 100$. We benchmark the library on three datasets across three group/domain pairs — PCam under C_8, OrganMNIST3D under the octahedral group, and Spherical MNIST under $\mathrm{SO}(3)$ — comparing against norm pooling, gated pooling, Fourier-ELU pooling, max pooling, and (non-equivariant) augmented CNN baselines. The library delivers near-exact rotation invariance (test-set rotation-accuracy std $\sigma_{\mathrm{rot}} < 10^{-3}$) and a data-efficiency advantage over incomplete alternatives in the low-data, moderate-capacity regime; the selective forward pass runs in sub-millisecond time on GPU for finite groups. Code is available at `https://anonymous.4open.science/r/bispectrum/`.

1 Introduction

Many machine-learning tasks are invariant under the action of a symmetry group G: histopathology classification under planar rotations C_n, signal classification under translations, volumetric organ classification under cube rotations O, and spherical-image classification under 3D rotations $\mathrm{SO}(3)$. Equivariant neural networks [Cohen and Welling, 2016, Weiler and Cesa, 2019] enforce these symmetries by construction in their feature maps, but at the point where a prediction must be made any task that does not depend on the group action — classification, regression, detection — still has to map the equivariant features to an *invariant* representation.

The *G-bispectrum* is a complex-valued vector of triple-correlation coefficients of the group Fourier transform. It has three properties that matter for ML: (i) **completeness** — it preserves all signal information up to the group action (the map is one-to-one on group orbits); (ii) **inversion** — the signal can in principle be recovered from the invariant up to its orbit; and (iii) **selectivity** — a

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small $O(|G|)$ index subset of the $O(|G|^2)$ full bispectrum still suffices for completeness on generic signals [Mataigne et al., 2024].

Existing invariant pooling layers — norm pooling, gate-based pooling, max/sum aggregation [Weiler and Cesa, 2019] — are simple but *incomplete*: they collapse equivariant features to scalars and discard information that distinguishes signals in different orbits. The bispectrum offers a mathematically principled alternative [Giannakis, 1989, Kakarala, 2012] and has been used as a competitive pooling layer in equivariant networks [Sanborn et al., 2023, Mataigne et al., 2024].

Despite this, bispectral pooling has not been widely adopted, for both theoretical and computational reasons. Existing bispectral constructions are scattered across theory papers, domain-specific derivations, and small implementations, with no general-purpose PyTorch library covering the discrete and continuous symmetry groups relevant to modern geometric deep learning. Even for groups where selectivity [Mataigne et al., 2024] reduces the bispectrum from $O(|G|^2)$ to $O(|G|)$ scalars (and the per-pass cost from $O(|G|^3)$ to $O(|G|^2)$), a user still needs a numerically stable implementation, a consistent API, an inversion routine, tests, and benchmarked runtime. On the theoretical side, the selective construction of Mataigne et al. [2024] applies to finite groups; the continuous case of $\mathrm{SO}(3)$ acting on the sphere S^2 has remained open.

This paper presents `bispectrum`, a PyTorch [Paszke et al., 2019] library for computing bispectral invariants and, where available, their inverses. The library supports the C_n, D_n, $\mathbb{T}^d$, disk, S^2, and octahedral domains as differentiable `nn.Modules` intended to plug into standard ML workflows. Beyond the software contribution, the paper makes one technical contribution that arose during implementation: a selective $\mathrm{SO}(3)$ bispectrum on S^2 with $\Theta(L^2)$ coefficients (Section 6).

Contributions.

1. **Software contribution.** A PyTorch library implementing selective G-bispectra for *seven different group/domain pairs* (a *domain* is the underlying space on which the group acts: C_n on C_n, $\mathrm{SO}(2)$ on S^1, $\mathbb{T}^d$, D_n, $\mathrm{SO}(2)$ on the disk, $\mathrm{SO}(3)$ on S^2, and the octahedral group O) as drop-in differentiable `nn.Modules` with a consistent API, GPU-optimized forward passes ($<$1 ms for finite groups), precomputed CG buffers, and a fully unit-tested codebase (96.8% line coverage).
2. **Theoretical contribution: selective $\mathrm{SO}(3)$ bispectrum on S^2.** We construct an invariant on S^2 that uses $\Theta(L^2)$ scalar coefficients (matching the orbit-space dimension $(L+1)^2 - 3$ up to a constant factor) instead of the $O(L^3)$ of the full bispectrum. We give a Kakarala-style *conditional* completeness theorem reducing the question to two named hypotheses — finite-band seed injectivity (H1, motivated by Kakarala [1992, 2012] and Bandeira et al. [2017]) and uniform bootstrap-determinant nonvanishing (H2) — both supported by per-degree numerical witnesses at a deterministic seed-42 real-signal point for every $\ell \leq 100$; the verification script is uniform in L and extends to larger band-limits without changes to the structural argument (Section 6, Appendix A).
3. **Empirical validation.** A benchmark of G-bispectral pooling across three ML tasks — PCam under C_8, OrganMNIST3D under the octahedral group O, and Spherical MNIST under $\mathrm{SO}(3)$ — against norm pooling, gated pooling, Fourier-ELU pooling, max pooling, and (non-equivariant) augmented-CNN baselines. Complete bispectral invariants provide a data-efficiency advantage in the low-data, moderate-capacity regime while converging to incomplete alternatives at high capacity.

Together, these contributions provide the first general-purpose, drop-in differentiable library for G-bispectral invariants and the first practical extension of the selective construction from finite groups to $\mathrm{SO}(3)$ on S^2.

2 Mathematical Background

Notation. A *group* G is a set with an associative product, identity, and inverse (e.g. rotations of the plane, translations of an image grid). G is *abelian* if elements commute (e.g. cyclic groups C_n, the torus $\mathbb{T}^d$) and *non-abelian* otherwise (e.g. $\mathrm{SO}(3)$, the dihedral groups D_n, the octahedral group O). A *representation* ρ assigns to each $g \in G$ a unitary matrix $\rho(g)$ such that $\rho(gh) = \rho(g)\rho(h)$; an

Table 1: Invariant nonlinearities for equivariant CNNs. "Exact"* means the invariant has no discretisation/aliasing error; "Complete" means it separates orbits; the coefficient column counts the size of the output for a signal with $|G|$ Fourier samples. "Genericity assumption"† means completeness holds outside a Lebesgue-measure-zero algebraic subvariety of the signal space. The selective bispectrum is the only method that is simultaneously exact, complete, and linear-cost for finite groups.

Nonlinearity	Exact?*	Complete?	Coefficients	Forward cost	Key limitation
Norm	Yes	No	$O(\|G\|)$	$O(\|G\|)$	Discards phase
Gate	Yes	No	$O(\|G\|)$	$O(\|G\|)$	Scalar bottleneck
Fourier pointwise	No	Yes‡	$O(\|G\|)$	$O(\|G\| \log \|G\|)$	Aliasing
Full bispectrum	Yes	Yes	$O(\|G\|^2)$	$O(\|G\|^3)$	Quadratic cost
Selective bispectrum	**Yes**	**Yes**	$\mathbf{O(\|G\|)}$	$\mathbf{O(\|G\|^2)}$	Genericity assumption†

‡Complete pre-aliasing; the pointwise nonlinearity in the forward pass introduces aliasing that may break completeness in practice.

irreducible representation (*irrep*) admits no nontrivial invariant subspace. The *dual* $\hat{G}$ denotes the set of equivalence classes of irreps; for finite G, $|\hat{G}| = O(|G|)$. A *group action* of G on a domain X is a map $G \times X \to X$, $(g, x) \mapsto g \cdot x$, compatible with the group product; it induces an action on functions $f : X \to \mathbb{R}$ by $(f \circ g)(x) := f(g \cdot x)$. Two signals are in the same *orbit* if one is the image of the other under some $g \in G$; an *invariant* of f is a function constant on each orbit. For compact G there is a unique normalised translation-invariant measure $d\mu$ called the *Haar measure*; finite G is the special case where $d\mu$ is the counting measure. "*Lebesgue-almost-every* f" means "every f outside a measure-zero algebraic subvariety of the signal space." The groups and domains used in this paper are: the cyclic group C_n (discrete planar rotations), the dihedral group D_n (rotations and reflections of a regular n-gon), the torus $\mathbb{T}^d = (S^1)^d$ (multi-channel periodic signals), $\mathrm{SO}(2)$ acting on the disk (continuous planar rotations of a 2D image), $\mathrm{SO}(3)$ acting on the sphere S^2 (continuous 3D rotations of a spherical signal), and the chiral octahedral group O (24 proper rotations of the cube, acting on 3D voxel grids).

2.1 Equivariant features and invariant maps

A neural network layer Φ is *G-equivariant* if its output transforms predictably under the group action: $[\pi_{\mathrm{out}}(g) \circ \Phi](f) = \Phi(\pi_{\mathrm{in}}(g)f)$ for all $g \in G$ [Cohen and Welling, 2016, Weiler and Cesa, 2019]. Here f is an input signal (a function on the domain X), $\pi_{\mathrm{in}}, \pi_{\mathrm{out}}$ are the (representations of the) G-actions on the input and output feature spaces of Φ, and the composition $\pi_{\mathrm{out}}(g) \circ \Phi$ is the action of g on the output side. Stacking equivariant layers preserves equivariance, but any task that requires a prediction independent of the group action—classification, regression, detection—must eventually map equivariant features to *invariant* ones (constant on each orbit).

Table 1 summarises the trade-offs between exactness, completeness, and computational cost across existing invariant pooling layers. Norm pooling and gated nonlinearities are exact invariants but *incomplete*: they discard the phase of Fourier blocks, collapsing distinct orbits. Fourier-space pointwise nonlinearities preserve completeness but sacrifice exactness through aliasing [Weiler and Cesa, 2019]. The full bispectrum is exact and complete but costs $O(|G|^2)$ coefficients. The *selective* bispectrum (introduced below) preserves both exactness and completeness while using only $O(|G|)$ coefficients on generic signals.

2.2 The G-bispectrum and selective bispectra

Let G be a compact group with dual $\hat{G}$ and irreps $\{\rho_i\}_{i \in \hat{G}}$ of dimensions d_i. The *group Fourier transform* [Diaconis and Rockmore, 1990] of a signal $f : G \to \mathbb{C}$ is

$$\hat{f}(\rho_i) = \int_G f(g)\, \rho_i(g)\, d\mu(g) \quad \in \mathbb{C}^{d_i \times d_i}, \tag{1}$$

which specialises to $\sum_{g \in G} f(g)\, \rho_i(g)$ when G is finite (counting measure). The Clebsch–Gordan (CG) decomposition $\rho_i \otimes \rho_j \cong \bigoplus_{k \in \hat{G}} m_{ij}^k \rho_k$ expresses the tensor product of two irreps as a direct sum of irreps (with integer multiplicities $m_{ij}^k \geq 0$); the change-of-basis is realised by the *CG isom-*

etry $C_{ij}^k : V_{\rho_k} \to V_{\rho_i} \otimes V_{\rho_j}$, projecting the tensor product onto its ρ_k-isotypic component. For each admissible triple (i, j, k) with $m_{ij}^k > 0$, the *bispectrum* is the scalar

$$\beta_{i,j}^k(f) := \left\langle \, (C_{ij}^k)^\dagger \left(\hat{f}(\rho_i) \otimes \hat{f}(\rho_j)\right), \, \hat{f}(\rho_k) \right\rangle. \tag{2}$$

Each $\beta_{i,j}^k$ is invariant under the action $f \mapsto f \circ g$ and, taken over all admissible triples, the bispectrum $\{\beta_{i,j}^k\}_{(i,j,k)}$ is a *complete* invariant: it separates orbits for generic signals [Kakarala, 2012]. For finite G, $|\hat{G}| = O(|G|)$, so the full bispectrum has $O(|G|^2)$ scalar coefficients (and is naturally indexed by $(i, j) \in \hat{G}^2$ with k ranging over the CG-allowed irreps). Equation (2) is stated for G acting on itself; for homogeneous spaces of the form G/H (e.g. $\mathrm{SO}(3)$ on S^2, or $\mathrm{SO}(2)$ on the disk), the same construction applies after expanding f in a basis adapted to G/H (spherical harmonics for S^2, Bessel modes on the disk). The explicit specialisation for S^2 is in Definition 5.

Scalar vs. matrix (vector) bispectrum. Throughout this paper the term *scalar bispectrum* refers to the single complex number $\beta_{i,j}^k(f)$ obtained from the inner product in Eq. (2)—i.e., the CG-contracted scalar associated with each admissible triple. In the literature one also encounters the *matrix* or *vector* bispectrum, which keeps the un-contracted matrix $(C_{ij}^k)^\dagger(\hat{f}(\rho_i) \otimes \hat{f}(\rho_j))$ before pairing it with $\hat{f}(\rho_k)$. The two are related by a single trace/inner-product step; the scalar form is what the library outputs and what we analyse for $\mathrm{SO}(3)$ on S^2.

Selective bispectra. Mataigne et al. [2024] show that for finite groups, a breadth-first search (BFS) over the Kronecker product table of $\hat{G}$ selects $O(|G|)$ bispectral coefficients $\beta_{i,j}^k$ that remain complete; Myers and Miolane [2025] extend this to the disk. This reduction from quadratic to linear is the key to practical deployment: for example, the bispectrum of the octahedral group drops from 576 to 172 coefficients.

3 Related Work

Equivariant network libraries. `escnn` [Cesa et al., 2022] and `e3nn` [Geiger and Smidt, 2022] provide general-purpose equivariant convolution layers for 2D and 3D groups respectively, and `lie-nn` [Batatia et al., 2023] provides general Lie-group representation primitives (irreducible representations, Clebsch–Gordan decompositions, tensor products) on top of which custom equivariant layers can be built. These libraries implement equivariant *feature maps* but leave the choice of invariant pooling to the user (typically norm or gate). Our library is complementary: it provides the invariant map as a drop-in `nn.Module` that plugs into architectures built with any of these frameworks.

Bispectral approaches in neural networks. Sanborn et al. [2023] introduced bispectral neural networks (BNNs), using the full G-bispectrum as an invariant nonlinearity for C_n and D_n. Oreiller et al. [2022] applied bispectral CNNs to medical imaging with a fixed set of hand-selected triples. Both use the full (quadratic-cost) bispectrum and provide implementations tied to specific architectures. Our library generalizes to seven group/domain pairs, defaults to the $O(|G|)$ selective reduction [Mataigne et al., 2024], and exposes a framework-agnostic API. Chevalley et al. [2024] explore bispectral signatures for data characterisation, using bispectra as descriptors rather than learnable pooling primitives. Outside the equivariant-network setting, `PyBispectra` [Binns et al., 2025] computes bispectrum and bicoherence statistics of one-dimensional time-series signals for electrophysiology applications such as phase-amplitude coupling and time-delay estimation; the underlying 1D bispectrum is translation-invariant by construction, but the toolbox is not designed as a neural-network pooling layer.

Bispectra on homogeneous and non-homogeneous spaces. The selective G-bispectrum construction [Mataigne et al., 2024] reduces the cost of the G-bispectrum from $O(|G|^2)$ to $O(|G|)$ scalars on finite groups. Myers and Miolane [2025] extend the selective construction to the disk, where $\mathrm{SO}(2)$ acts on a non-homogeneous space. Kakarala [2012] treats the $\mathrm{SO}(3)$-on-$\mathrm{SO}(3)$ bispectrum and the closely related sphere case via Wigner-D irreps. The matrix (vector) bispectrum on S^2 is the homogeneous-space form on which our scalar contractions and CG power augmentation are built.

Table 2: Library profiling summary. Timings are median wall-clock on a single NVIDIA H100 80 GB GPU. "Coefs" = number of scalar coefficients (Sel. = selective, Full = full bispectrum); "Fwd" = forward-pass time in milliseconds; "Inv" = inversion time in milliseconds. "–" = not implemented. For continuous groups, $|G| = \infty$; coefficient counts refer to the band-limited representation at the tested resolution. *`SO3onS2` benchmarked at band-limit $\ell_{\max}$=16 (we use $\ell_{\max}$ and L interchangeably to denote the band-limit; see Section 6).

Module	G	$\|G\|$	Coefs		Fwd (ms)		Inv (ms)
			Sel.	Full	Sel.	Full	Sel.
`CnonCn`	C_{128}	128	128	8256	0.15	9.17	8.56
`TorusOnTorus`	$C_{32}\times C_{32}$	1024	1024	524800	0.08	0.71	68.02
`DnonDn`	D_{32}	64	245	–	2.52	–	42.33
`SO2onDisk`	$\mathrm{SO}(2)$	∞	411	–	1.94	–	17.69
`SO3onS2`*	$\mathrm{SO}(3)$	∞	430	–	26.54	–	–
`OctaonOcta`	O	24	172	–	0.70	–	–

CG product nonlinearities and tensor products. Tensor field networks [Thomas et al., 2018] and MACE [Batatia et al., 2022] use iterated Clebsch–Gordan products to build equivariant interactions. These products are exact and complete in principle, but each layer incurs $O(L^3)$ cost per CG contraction. Xie et al. [2025] analyse the *price of freedom* in tensor product architectures, showing that maintaining full expressivity through CG products becomes expensive at high angular momenta. The selective bispectrum achieves completeness (conditionally on H1 and H2 for $\mathrm{SO}(3)$ on S^2, see Section 6) in a single $O(L^2)$ pass for $\mathrm{SO}(3)$ or $O(|G|)$ pass for finite groups, avoiding iterated contractions.

Invariant theory and scattering. The wavelet scattering transform [Mallat, 2012] provides translation-invariant representations via iterated modulus operations; Esteves et al. [2018] extend this to the sphere. Unlike the G-bispectrum, scattering representations are stable but generally incomplete—they discard inter-scale phase information. The G-bispectrum's completeness guarantee [Giannakis, 1989, Kakarala, 2012] is its distinguishing theoretical advantage.

4 The `bispectrum` Library

4.1 Design and implementation

Every group/domain pair is exposed as a `torch.nn.Module` named `{Group}on{Domain}` (e.g. `SO3onS2`). Each module exposes a uniform API: `forward(f)` returns the selective bispectral invariants of f; `fourier(f)` returns the underlying group Fourier coefficients; and, where available, `invert(beta)` returns the reconstructed signal up to group-action indeterminacy. Modules accept raw real-valued inputs, handle the Fourier transform internally, and default to $O(|G|)$ selective coefficients (`selective=False` gives the full $O(|G|^2)$ set). Clebsch–Gordan (CG) matrices, discrete Fourier transform (DFT) kernels, and Bessel roots (used for the radial discretisation of the disk in `SO2onDisk`) are precomputed at construction and stored as non-learnable buffers; this lets `.to(device)` migrate the module transparently and ensures that forward passes incur zero recomputation. The only dependencies are PyTorch [Paszke et al., 2019], NumPy [Harris et al., 2020], and `torch_harmonics` [Bonev et al., 2023] (for `SO3onS2`). For $\mathrm{SO}(3)$ on S^2, reduced CG matrices are built in parallel and cached to disk; all forward methods are batch-native, using `torch.einsum`/`torch.bmm` for non-abelian modules and `torch.fft` for abelian ones. The test suite covers 96.8% of lines across all modules. A minimal usage example is in Appendix G.

4.2 Profiling

We profile the library on a single NVIDIA H100 80 GB GPU (PyTorch 2.10, batch = 16 for the forward pass and 4 for inversion, median wall-clock via `torch.utils.benchmark`); results are in Table 2 and Figure 1.

The selective forward pass is sub-millisecond for finite abelian modules (C_n: 0.15 ms, $\mathbb{T}^2$: 0.08 ms), under 2 ms for $\mathrm{SO}(2)$/disk (1.94 ms), and under 1 ms for the octahedral group (0.70 ms), making it

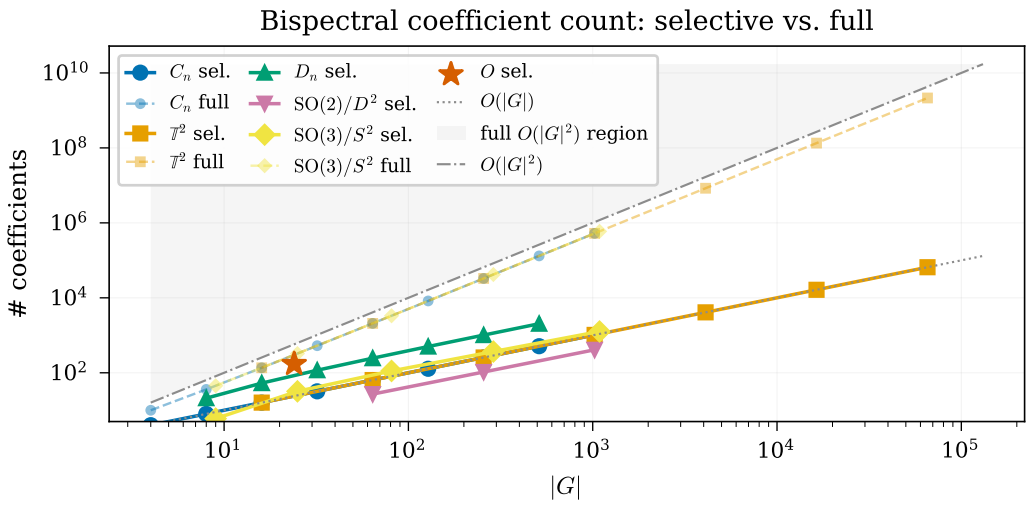

Figure 1: Bispectral coefficient count vs. group order $|G|$. Solid lines: selective ($O(|G|)$); dashed: full ($O(|G|^2)$). At $|G|$=1,024, selectivity yields a 512× reduction. Continuous groups (SO(2), SO(3)) are plotted at their band-limited signal dimension. The full-bispectrum curves for D_n and the disk are omitted because the corresponding backends are not yet implemented in the library.

negligible relative to backbone computation. SO(3) on S^2 at $\ell_{\max}$=16 takes 26.5 ms, dominated by the spherical harmonic transform (via `torch_harmonics` [Bonev et al., 2023]). Selective coefficient counts confirm the $O(|G|)$ scaling: C_{128} uses 128 coefficients vs. 8,256 full (64× reduction); $\mathbb{T}^2$ at $|G|$=1,024 uses 1,024 vs. 524,800 full (512×). Inversion is fast for abelian groups (9–68 ms). GPU throughput scales linearly with batch size, exceeding 10^7 samples/s for C_n and $\mathbb{T}^2$ (Appendix C).

5 Experiments

We validate the library on three tasks spanning discrete and continuous groups: 2D histopathology classification under C_8 symmetry (PCam, Section 5.1), 3D organ classification under octahedral symmetry (OrganMNIST3D, Section 5.2), and spherical digit classification under full SO(3) (Spherical MNIST, Section 5.3).

5.1 2D histopathology classification under cyclic symmetry

We evaluate five invariant pooling strategies on PatchCamelyon [Veeling et al., 2018] (PCam), 327K histopathology patches with binary metastasis labels. All equivariant models share a C_8-equivariant DenseNet [Huang et al., 2017] backbone following Veeling et al. [2018]; they differ only in the terminal invariant map: norm pooling, gated pooling, Fourier-ELU, and selective bispectrum via `CnonCn`(n=8). A non-equivariant standard CNN trained with data augmentation drawn uniformly from the symmetry group C_8 ("*augmented CNN*") serves as the non-equivariant baseline. We sweep growth rates to vary capacity (30K–1.6M params) and train on 10% of the PCam training set (an image-level subsample, not a group-orbit subsample, drawn before any group augmentation) with AdamW [Loshchilov and Hutter, 2019]. Full model descriptions are in Appendix D.

Results. Table 3 and Figure 2 present the results at matched ∼100K parameters.

At matched capacity (∼100K params, 10% data), all four equivariant pooling methods substantially outperform the standard augmented CNN (AUC 0.896). Fourier-ELU leads slightly (0.945), followed by NormReLU (0.942), bispectrum and gated (both 0.941). At 1% data, the bispectrum leads all methods (0.912), outperforming NormReLU (0.873) and Fourier-ELU (0.876). This data-efficiency advantage narrows as the training set grows, consistent with complete invariants providing a stronger inductive bias when data is scarce.

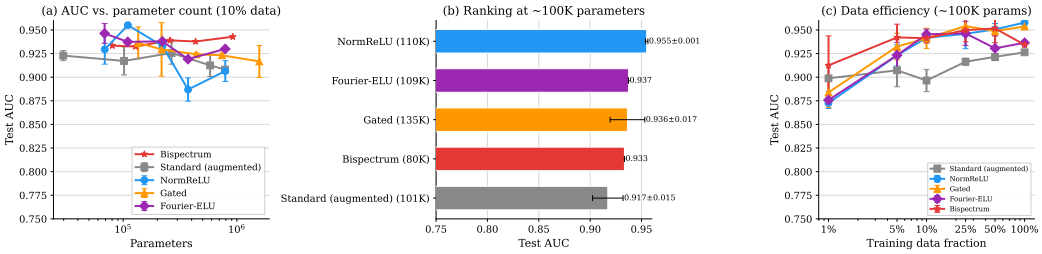

Figure 2: PCam Pareto analysis (`CnonCn` selective bispectrum, single seed, 10% training data). (a) Test AUC vs. parameter count across all five pooling variants. (b) Ranking at the ∼100K parameter budget. (c) Data efficiency at matched ∼100K parameters. Multi-seed statistics are in Table 3.

Table 3: PCam results at matched ∼100K parameters (10% training data, mean ± std over 3 seeds). All equivariant models share a C_8-equivariant DenseNet backbone.

Model	Params	Test AUC	Test ACC
Standard (augmented)	102K	.896 ± .010	.826 ± .012
Gated	136K	.941 ± .009	.870 ± .004
Selective Bispectrum (`CnonCn`)	128K	.941 ± .004	**.872 ± .006**
Fourier-ELU	110K	**.945 ± .004**	.855 ± .006
NormReLU	110K	.942 ± .004	.858 ± .007

5.2 3D organ classification under octahedral symmetry

To evaluate the `OctaonOcta` module for the chiral octahedral group O ($|O| = 24$), we design a controlled 3D experiment on OrganMNIST3D [Yang et al., 2023] (1,742 abdominal CT volumes, 28^3 voxels, 11 classes). All equivariant variants share an O-equivariant 3D ResNet backbone with exact voxel-permutation kernels. The four models compared are: a translation-equivariant standard 3D CNN trained with octahedral data augmentation (the "*augmented CNN*" baseline, 16K params); the same O-equivariant ResNet with terminal norm pool (375K params); with terminal max pool (374K params); and with terminal selective bispectrum via `OctaonOcta` (463K, 172 invariant scalars). Models are trained with AdamW [Loshchilov and Hutter, 2019] for 100 epochs with three seeds; rotation robustness is measured over all 24 octahedral rotations of the test set. Published baselines and wider-channel experiments are in Appendix E.

Results. Table 4 and Figure 3 report the key findings.

Rotation invariance. All equivariant models achieve $\sigma_{\text{rot}} \leq 0.0005$, while the augmented CNN drops ∼2.5 pp (percentage points) under rotation.

Completeness vs. lossy pooling. The bispectrum (72.6%) matches max pool (73.0%) and substantially outperforms norm pool (56.8%) and Regular SE(3) convolutions [Kuipers and Bekkers, 2023] (69.8%). At reduced data (Figure 3), the bispectrum's advantage widens: at 5% data it reaches 33.2% vs. 18.9% for max pool. At higher capacity (6M params, Appendix E), max pool overtakes the bispectrum, suggesting the completeness advantage is specific to the data-limited regime.

5.3 Spherical MNIST: SO(3) invariance on the sphere

The preceding experiments validate bispectral pooling for finite groups (C_8 and the octahedral group O). To evaluate the library's continuous group support, we test `SO3onS2`—the $\mathrm{SO}(3)$ bispectrum on the 2-sphere—on Spherical MNIST [Cohen et al., 2018], the standard benchmark for rotation invariance on S^2.

Setup. MNIST digits are projected onto the sphere via stereographic projection and sampled on a 64×128 equiangular grid. We compute the selective bispectrum at $\ell_{\max} = 15$, producing 384 com-

Table 4: OrganMNIST3D results (mean $\pm$ std over 3 seeds). Rot ACC and σ_{rot}: mean and std of accuracy across 24 octahedral test-set rotations.

Model	Params	Test ACC	Test AUC	Rot ACC	σ_{rot}
Standard 3D CNN	16K	$.601 \pm .017$	$.951 \pm .002$	.598	.0115
O-Equiv + Norm Pool	375K	$.568 \pm .099$	$.940 \pm .021$	.568	.0000
O-Equiv + Max Pool	374K	$.730 \pm .033$	$.972 \pm .005$	.730	.0000
O-Equiv + `OctaonOcta` Bispectrum	463K	$.726 \pm .027$	$.972 \pm .004$	.726	.0005†

†Nonzero σ_{rot} due to floating-point precision in CG contractions, not a theoretical invariance failure.

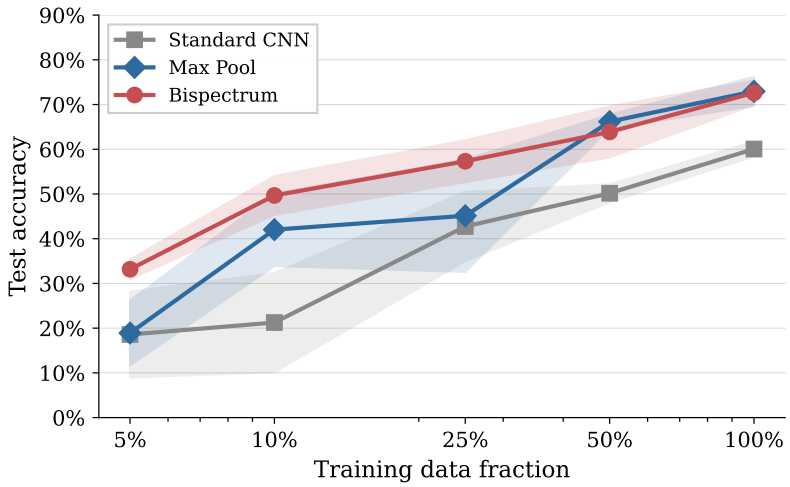

Figure 3: Test accuracy vs. training data fraction on OrganMNIST3D (`OctaonOcta` bispectrum, mean over 3 seeds; shaded bands: ± 1 std). The complete invariant map provides a consistent advantage in the low-data regime.

plex invariant coefficients. To form the network input we apply a sign-preserving $\mathrm{sign}(x)\log(1+|x|)$ transform separately to the real and imaginary parts of each coefficient, yielding 768 real features. By Lemma 28 exactly one of the two log1p features per coefficient is structurally zero on real signals (real for even parity, imaginary for odd parity), so the effective feature count is ≈ 384 reals; we keep both channels because (i) the transform is sign-preserving and per-channel injective, so the sign of $\mathrm{Im}(\beta_{2,3,4})$ that resolves the T_R ambiguity (Cor. 32) is preserved, and (ii) the $\log 1p$ non-linearity is part of the empirical model and is not used in the formal completeness theorem (Theorem 2). The features are fed to a 3-layer MLP (232K parameters total). The conditional completeness theorem (Appendix A) is supported by per-degree numerical witnesses for every $\ell \leq 100$, so $\ell_{\max} = 15$ is well within the verified range. Three baselines provide controls:

- **Power Spectrum + MLP** (11K params): computes only the rotation-invariant power spectrum $\|f_\ell\|^2$ per degree ℓ, discarding all phase coupling between harmonics. Invariant but *incomplete*.
- **Power Spectrum + MLP (matched)** (166K params): same incomplete invariant features but with a larger MLP (166K), isolating the effect of feature completeness from model capacity.
- **Standard CNN** (185K params): a conventional CNN operating directly on the equirectangular image. Not rotation-invariant.

Following Cohen et al. [2018], we evaluate under three protocols: train on non-rotated / test on non-rotated (NR/NR), train on rotated / test on rotated (R/R), and the most challenging train on non-rotated / test on rotated (NR/R). Each configuration is run with three seeds. Rotation robustness is assessed by applying 10 random $\mathrm{SO}(3)$ rotations to the entire test set and measuring accuracy variance.

Table 5: Spherical MNIST results (mean $\pm$ std over 3 seeds). Reference numbers are from Cohen et al. [2018]. σ_{rot}: standard deviation of accuracy under 10 random $\mathrm{SO}(3)$ rotations (NR-trained models).

Model	Params	NR/NR ↑	R/R ↑	NR/R ↑	σ_{rot} ↓
Cohen planar CNN	—	0.98	0.23	0.11	—
Cohen spherical CNN	—	0.96	0.95	0.94	—
Standard CNN	185K	0.991±0.000	0.781±0.003	0.138±0.003	0.044
Power Spec. + MLP	11K	0.780±0.002	0.780±0.002	0.777±0.001	3×10^{-4}
Power Spec. + MLP (matched)	166K	0.794±0.011	0.794±0.011	0.792±0.010	3×10^{-4}
`SO3onS2` **Bispectrum + MLP (ours)**	232K	**0.950±0.000**	**0.949±0.002**	**0.951±0.001**	$\mathbf{1\times10^{-4}}$

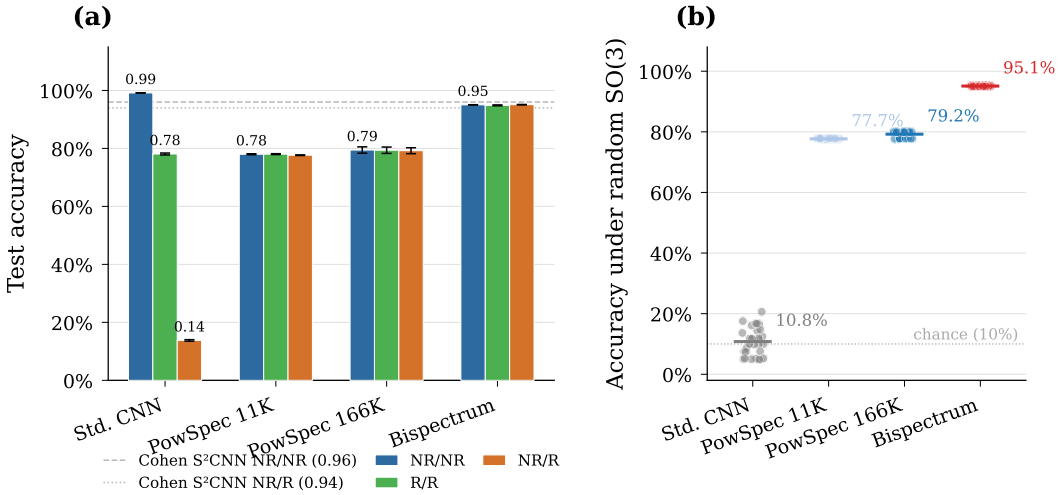

Figure 4: Spherical MNIST analysis (`SO3onS2` selective bispectrum). **(a)** Cohen et al. protocol: NR/NR, R/R, and NR/R accuracy. Dashed lines show Cohen's spherical CNN reference. **(b)** Accuracy under 10 random $\mathrm{SO}(3)$ rotations (NR-trained models, 30 points per model = 10 rotations $\times$ 3 seeds). Scaling the power spectrum MLP from 11K to 166K parameters yields only +1.4pp; the `SO3onS2` bispectrum at 232K reaches 95.0%.

Results. Table 5 and Figure 4 report the results.

Numerically exact $\mathrm{SO}(3)$ invariance. The bispectrum achieves near-identical accuracy across all three protocols ($\sigma_{\text{rot}} \sim 10^{-4}$), confirming true $\mathrm{SO}(3)$ invariance. The standard CNN collapses from 99.1% (NR/NR) to 13.8% (NR/R), showing that augmentation alone does not confer invariance.

Completeness gap. The power spectrum is equally invariant but reaches only 78%—a 17pp deficit that persists even after scaling the MLP to 166K parameters (+1.4pp), confirming incompleteness rather than capacity as the bottleneck.

Comparison with Cohen et al. Cohen's spherical CNN [Cohen et al., 2018] reaches 96%/95%/94% with learned equivariant convolutions. Our single-pass bispectrum (95.0%/94.9%/95.1%) is 1–2pp lower but has a tighter invariance gap (0.1pp vs. 2.0pp).

6 Selective $\mathrm{SO}(3)$ bispectrum on S^2

6.1 Setup

A real signal $f : S^2 \to \mathbb{R}$ band-limited at degree L expands in spherical harmonics as $f(\theta, \varphi) = \sum_{\ell=0}^{L} \sum_{m=-\ell}^{\ell} a_\ell^m Y_\ell^m(\theta, \varphi)$ with the reality constraint $a_\ell^{-m} = (-1)^m \overline{a_\ell^m}$, so it is parameterised

by $(L{+}1)^2$ real scalars (Definition 5, eq. (5)). On $S^2 = \mathrm{SO}(3)/\mathrm{SO}(2)_z$, the relevant irreducible representations of $\mathrm{SO}(3)$ are $\{\rho_\ell\}_{\ell\geq 0}$ with V_{ρ_ℓ} the $(2\ell{+}1)$-dimensional space of degree-ℓ spherical harmonics and Wigner-D action $\rho_\ell(g) = D^\ell(g)$. In the irrep notation of Mataigne et al. [2024], the Fourier coefficient at ρ_ℓ is $\mathcal{F}(f)_{\rho_\ell} = \mathbf{F}_\ell \in \mathbb{C}^{2\ell+1}$; we write $\mathbf{F}_\ell$ throughout. The scalar ρ_ℓ-isotypic component of the matrix-valued bispectrum $\beta_{\rho_{\ell_1},\rho_{\ell_2}}(f)$ is the familiar

$$\beta_{\ell_1,\ell_2,\ell} = \sum_{\substack{m_1,m_2,m \\ m_1+m_2=m}} C^{\ell,\,m}_{\ell_1,\,m_1;\,\ell_2,\,m_2}\, a^{m_1}_{\ell_1}\, a^{m_2}_{\ell_2}\, \overline{a^{m}_{\ell}}, \tag{3}$$

the specialisation of eq. (2) to $\mathrm{SO}(3)$-on-S^2. The full bispectrum has $\Theta(L^3)$ such scalars; the orbit space $\mathbb{R}^{(L+1)^2}/\mathrm{SO}(3)$ has dimension $(L{+}1)^2 - 3$, so any complete invariant requires $\Omega(L^2)$ independent components (Proposition 9). The Mataigne chain $\beta_{1,\ell,\ell+1}$ for $\ell = 0, \ldots, L{-}1$ provides only $\sim 3L$ scalars and cannot achieve this bound.

6.2 Selective construction and algorithm

Let $L_{\mathrm{seed}} := \min(L, 4)$. The *seed index set* collects all admissible bispectral triples up to L_{seed},

$$\mathcal{S}_{\mathrm{seed}} := \{(\ell_1,\ell_2,\ell) : 0 \leq \ell_1 \leq \ell_2 \leq L_{\mathrm{seed}},\ |\ell_1{-}\ell_2| \leq \ell \leq \min(\ell_1{+}\ell_2, L_{\mathrm{seed}})\},$$

of constant size ≤ 50 once $L \geq 4$, matching the implementation seed (Algorithm 2). For each high-degree target $\ell \geq 5$, we use a size-$(2\ell{+}1)$ *bootstrap block* $\mathcal{T}_\ell$, which for $\ell \geq 8$ is the closed-form family of eq. (8) (Definition 12) and for $5 \leq \ell \leq 7$ is the explicit small block $\mathcal{T}^{\mathrm{small}}_\ell$ tabulated in Algorithm 2. Both versions consist of entries linear in $\mathbf{F}_\ell$ with coefficients depending only on already-recovered lower degrees. The *selective* $\mathrm{SO}(3)$ *bispectrum on* S^2 is then the concatenation

$$\beta_{\mathrm{sel}}(f) := \left\{\beta_{\ell_1,\ell_2,\ell}\right\}_{(\ell_1,\ell_2,\ell)\in\mathcal{S}_\beta}, \qquad \mathcal{S}_\beta := \mathcal{S}_{\mathrm{seed}} \cup \bigcup_{\ell=5}^{L} \mathcal{T}_\ell. \tag{4}$$

By Lemma 28, each scalar entry contributes exactly one real degree of freedom on the real-signal slice (real for even parity, purely imaginary for odd parity); we record that single real number per entry. Algorithm 1 writes this as a Mataigne-style BFS over selected triples; the implementation-grade index set (Algorithm 2 in Appendix A) augments $\mathcal{S}_\beta$ with mandatory even self-couplings and optional CG-power scalars for numerical robustness, and is a strict superset of β_{sel}.

Algorithm 1 Selective $\mathrm{SO}(3)$ bispectrum on S^2 (BFS form, mirroring Mataigne et al. [2024], Alg. 1).

```
Input: band-limit L, seed cutoff L_seed = min(L, 4)
L_β ← all admissible triples (ℓ1, ℓ2, ℓ) with ℓ ≤ L_seed {seed block S_seed}
for ℓ = L_seed+1, ..., L do
    append the bootstrap block T_ℓ to L_β {2ℓ+1 entries linear in F_ℓ; T_ℓ^small for ℓ ≤ 7, eq. (8) for ℓ ≥ 8}
end for
return β_sel(f) = {β_{ℓ1,ℓ2,ℓ}(f) : (ℓ1, ℓ2, ℓ) ∈ L_β}
```

The output size is $|\beta_{\mathrm{sel}}| = |\mathcal{S}_{\mathrm{seed}}| + \sum_{\ell=5}^{L}(2\ell{+}1) = \Theta(L^2)$, matching the orbit-space lower bound up to a constant (Proposition 14).

Remark 1 (CG power: needed in implementation, optional in formal statement). The library appends the CG power spectrum $P_{\ell_1,\ell_2,\ell} := \|(\mathbf{F}_{\ell_1} \otimes \mathbf{F}_{\ell_2})|_\ell\|^2$ (Definition 6) at $O(L^2)$ extra cost. In the formal theorem we set $\mathcal{S}_P = \varnothing$ for the *stated* bispectral invariant β_{sel}; the proof of completeness then routes through Hypothesis H2, which is phrased over the augmented per-degree block (linear $\mathcal{T}_\ell$ plus quadratic/cubic self-couplings plus CG-power scalars). H2 is needed precisely because the pure $\mathcal{T}_\ell$ submatrix is rank-deficient on the real-signal slice (see Remark 35). The implementation includes the CG power scalars unconditionally for this reason and as a per-degree numerical robustness diagnostic.

6.3 Conditional completeness theorem

Status of the theory. What is *proved unconditionally* in Appendix A: the gauge-fixing reduction (Lemma 19); the closed-form recovery of $\mathbf{F}_0, \mathbf{F}_1$ from $\beta_{0,0,0}$ and $\beta_{0,1,1}$ (Lemma 21); the parity

Table 6: Theorem–implementation consistency at $L = 15$ and $L = 16$. Formal $|\beta_{\mathrm{sel}}|$ counts one real component per scalar entry (Lemma 28); the implementation `output_size` from `_build_selective_index_map` $\cup$ `_build_cg_power_index_map` is a strict superset, adding mandatory even self-couplings and CG-power scalars for numerical robustness; the Spherical MNIST feature vector applies a sign-preserving $\log 1p$ separately to the real and imaginary parts of each implementation scalar (Section 6).

L	Formal $\lvert\beta_{\mathrm{sel}}\rvert$	Impl. `output_size`	SMNIST features
15	273	384	768
16	306	430	860

structure (Lemma 28, Cor. 32, Prop. 30); and the bootstrap row count $|\mathcal{T}_\ell| = 2\ell+1$ (Prop. 13). What is *assumed*: H1 (finite seed injectivity at $L_{\mathrm{seed}} = 4$ for the scalar bispectrum) and H2 (uniform per-degree augmented Jacobian nonvanishing for $\ell \geq 5$). What is *numerical-witness only*: the H2 evidence at the deterministic seed-42 real-signal point for every $\ell \leq 100$, plus the empirical sub-claims of the constructive seed solver of Appendix B (degree-4 NLLS filter, T_R-branch inconsistency).

Theorem 2 reduces completeness of β_{sel} to two named hypotheses; both are stated formally in Appendix A.

H1 (finite seed injectivity, Hypothesis 25). The full *scalar* bispectrum at $L_{\mathrm{seed}} = 4$ is injective on a Zariski-open gauge-fixed stratum of $V_{L_{\mathrm{seed}}}$, with T_R resolved by the odd-parity entry $\mathrm{Im}(\beta_{2,3,4})$. Structural motivation: Kakarala's Theorem 7 [Kakarala, 1992, 2012] establishes injectivity for the *matrix* bispectrum on $L^1(\mathrm{SO}(3))$; H1 is the corresponding statement for the CG-contracted scalar map and is therefore a hypothesis, not a corollary of Theorem 7 (see Paragraph A.4.1 for the precise gap). Independent exact-arithmetic Gröbner-basis certificates of Bandeira et al. [2017] verify the scalar statement at $L_0 \leq 15$; no closed-form proof is known beyond their finite-degree range.

H2 (uniform bootstrap-determinant nonvanishing, Hypothesis 34). For every $\ell \geq 5$, the per-degree linear map from $\mathbf{F}_\ell$ to the augmented selective entries (the bootstrap rows $\mathcal{T}_\ell$ together with the mandatory even self-couplings and CG-power scalars at degree ℓ) is injective on a Zariski-open subset of the real-signal lower-degree space. The pure linear-bootstrap submatrix $A_\ell \in \mathbb{R}^{(2\ell+1)\times(2\ell+1)}$ assembled from $\mathcal{T}_\ell$ alone is rank-deficient on the real-signal slice; full per-degree rank is obtained only after appending the additional augmenting entries generated by Algorithm 2.

Theorem 2 (Conditional completeness of β_{sel})**.** *Let $L \geq 4$ and assume H1 and, if $L \geq 5$, H2. For $f, f' : S^2 \to \mathbb{R}$ band-limited at degree L and lying outside a proper algebraic subvariety of V_L (Assumption 16), if $\beta_{\mathrm{sel}}(f) = \beta_{\mathrm{sel}}(f')$, then $f' = g \cdot f$ for some $g \in \mathrm{SO}(3)$. The output β_{sel} has $\Theta(L^2)$ real components, matching the orbit-space dimension $(L+1)^2 - 3$ up to a constant factor.*

Inversion sketch (four steps). (i) *Gauge-fix* $\mathbf{F}_1$ along $\hat{z}$ and the residual $\mathrm{SO}(2)_z$ phase via $\arg a_2^1$, consuming all three rotational DOFs (Lemma 19). (ii) *Closed form for* $\mathbf{F}_0, \mathbf{F}_1$: $\beta_{0,0,0} = (a_0^0)^3$ pins A_0, then $\beta_{0,1,1} = A_0\|\mathbf{F}_1\|^2$ pins $\mathbf{F}_1 = (0, c, 0)^\top$ (Lemma 21). (iii) *Seed solve at* $\ell \leq L_{\mathrm{seed}} = 4$: under H1, the seed block $\mathcal{S}_{\mathrm{seed}}$ uniquely determines $(\mathbf{F}_2, \mathbf{F}_3, \mathbf{F}_4)$ on the gauge slice; the parity entry $\mathrm{Im}(\beta_{2,3,4})$ separates f from $T_R \cdot f$ (Corollary 32; on the "up-to-reflection" caveat of Edidin and Satriano [2024] see Remark 26). (iv) *Augmented per-degree solve for* $\ell \geq 5$: under H2, the augmented per-degree block (bootstrap rows $\mathcal{T}_\ell$ plus the mandatory even self-couplings and CG-power scalars at degree ℓ) determines $\mathbf{F}_\ell$ from $\mathbf{F}_{<\ell}$ on a Zariski-open subset of the lower-degree real-signal coordinates, and induction recovers $\mathbf{F}_5, \ldots, \mathbf{F}_L$.

6.4 Computational evidence and scope

Hypothesis H2 is checked numerically per degree at the deterministic seed-42 real-signal witness in IEEE 754 double precision: for every $\ell \in \{5, \ldots, 100\}$, the smallest singular value of the augmented per-degree block (bootstrap rows $\mathcal{T}_\ell$ together with the mandatory even self-couplings and CG-power scalars at degree ℓ) is well above the SVD numerical noise floor, with relative ratio $\sigma_{\min}/\sigma_{\max} \geq 10^{-4}$. The pure linear-bootstrap submatrix A_ℓ alone is rank-deficient on the real-signal slice; the augmenting entries are essential for the per-degree solve. Scripts: `paper/exact_certificates.py`

and `benchmarks/verify_linear_bootstrap.py`. This is a numerical *witness* at one rational floating-point point per degree, not a closed-form proof of H2 (interval arithmetic, an exact rational witness, or a closed-form Wigner-3j identity would upgrade H2 to unconditional). The structural form of $\mathcal{T}_\ell$ is uniform in $\ell \geq 8$, so extending the verified range beyond $L = 100$ requires only re-running the script with a larger upper bound. Hypothesis H1 is supported by Kakarala Theorem 7 [Kakarala, 2012, 1992] as structural motivation and by the exact-arithmetic certificates of Bandeira et al. [2017] for $L_0 \leq 15$, but neither alone constitutes a band-limited proof of H1. Three routes would upgrade H2 to an unconditional theorem: interval arithmetic at the witness, exact rational determinants per ℓ, or a closed-form Wigner-3j non-vanishing identity. Full proofs and the complete set of hypotheses are in Appendix A.

7 Discussion and Conclusion

We have presented `bispectrum`, a PyTorch library that makes selective G-bispectral invariants practical for machine learning. The library covers seven group/domain pairs with a uniform `nn.Module` interface and required one technical contribution during implementation: a selective $\mathrm{SO}(3)$ bispectrum on S^2 achieving the optimal $\Theta(L^2)$ coefficient count (Theorem 2). Table 2 confirms that the selective forward pass adds negligible overhead to training pipelines (<1 ms for all finite-group modules on a single GPU). Across three experimental settings, bispectral pooling provides numerically exact rotation invariance (note that other pooling strategies, such as norm or max pooling, also yield exact invariance; the bispectrum's distinguishing property is conditional completeness, not invariance per se) and a data-efficiency advantage over incomplete alternatives in the low-data / moderate-capacity regime, where the conditionally complete invariant map (Theorem 2 under Hypotheses H1 and H2) provides a stronger inductive bias that partially compensates for limited training data.

When not to use the bispectrum. The data-efficiency advantage of complete invariants is most pronounced in the low-data, moderate-capacity regime. At high model capacity (Section 5), incomplete alternatives such as max pooling can match or surpass bispectral pooling, likely because the network learns to compensate for missing invariant information. Each bispectrum module is also invariant to a *specific* group (e.g. rotations only, or translations only); for tasks that need invariance to multiple or unknown symmetries, learned invariance via data augmentation or group-equivariant networks with flexible symmetry may be a better fit. Completeness is a property of the *map*, not of downstream task performance: a complete invariant provides a stronger inductive bias, but does not guarantee superior accuracy on every task. Invertibility of the bispectrum for generic signals was first proved by Giannakis [1989]; Kakarala [2012] extended this to compact groups. For $\mathrm{SO}(3)$ on S^2, computational cost grows as $O(L^2)$ per forward pass; at very large band-limits this may become a bottleneck relative to simpler pooling strategies.

Limitations. Inversion for $\mathrm{SO}(3)$ on S^2 is not yet implemented; only the selective forward map is available. The selective $\mathrm{SO}(3)$ completeness statement is *conditional*: it depends on Hypothesis H1 (finite-band seed injectivity, motivated by Kakarala Theorem 7 [Kakarala, 1992, 2012] for the matrix bispectrum and the exact-arithmetic Gröbner-basis certificates of Bandeira et al. [2017] for $L_0 \leq 15$) and Hypothesis H2 (uniform per-degree augmented Jacobian nonvanishing). H2 is supported per ℓ by a numerical witness at the deterministic seed-42 real-signal point: the smallest singular value of the augmented per-degree Jacobian satisfies $\sigma_{\min}/\sigma_{\max} \geq 10^{-4}$ in double precision for every $\ell \leq 100$, well above the SVD numerical noise floor (this is consistent with non-degeneracy at the witness, but is not a closed-form proof). The constant-size seed Jacobian rank (Lemma 42) is verified in exact rational arithmetic via `sympy`. Upgrading H2 to an unconditional theorem requires interval arithmetic, exact rational Jacobian rank per ℓ, or a closed-form Wigner-3j non-vanishing identity; a full characterisation of the degenerate set is open. Our experimental comparisons do not include tensor-product architectures (e3nn, MACE) or wavelet scattering, which target different input assumptions and would require non-trivial adaptation to the benchmarks used here.

Future work. Key directions include: implementing $\mathrm{SO}(3)$ inversion from selective bispectral coefficients; a structural proof of bootstrap rank for all L; numerical conditioning analysis at large band-limits; broader framework integration (the library is intentionally framework-agnostic and already drops in as an invariant module for architectures built on `escnn` [Cesa et al., 2022] or `e3nn` [Geiger and Smidt, 2022]); and applications to cryo-EM reconstruction and molecular property prediction.

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A Selective $\mathrm{SO}(3)$ Bispectrum – Full Proof

This appendix proves the conditional completeness theorem of Section 6. The two load-bearing assumptions H1 (finite seed injectivity) and H2 (uniform bootstrap-determinant nonvanishing) named in the main text are stated formally below as Hypotheses 25 and 34. The proof follows the Kakarala pattern [Kakarala, 1992, 2012]: a constant-size full-bispectrum seed at degree $L_{\mathrm{seed}} \leq 7$ followed by a triangular per-degree linear recurrence in which each $\mathbf{F}_\ell$ ($\ell \geq 8$) is recovered by solving an explicit $(2\ell{+}1)$-dimensional system. Clebsch–Gordan power entries are optional implementation diagnostics and are not used in the proof ($\mathcal{S}_P = \varnothing$ throughout). The implementation-grade selective-index-set construction (Algorithm 2) is preserved at the end of Section A.3.

Table 7: Asymptotic entry counts in β_{sel}. The constant-size seed block $\mathcal{S}_{\mathrm{seed}}$ collects every triple at degrees $\leq L_{\mathrm{seed}} = \min(L, 7)$; the high-degree selective part allocates exactly $2\ell{+}1$ rows per target degree (the bootstrap block $\mathcal{T}_\ell$ of eq. (8)).

degree range	selected β	total
$\ell \leq 7$	full seed, ≤ 512	$O(1)$
$\ell \geq 8$	$2\ell{+}1$	$O(\ell)$

The total invariant size is therefore $|\mathcal{S}_{\mathrm{seed}}| + \sum_{\ell=8}^{L}(2\ell{+}1) = \Theta(L^2)$, matching the orbit-space lower bound $(L{+}1)^2 - 3$ of Proposition 9 up to a constant factor (Proposition 14).

A.1 Setup and notation

Definition 3 (Band-limited signal on S^2)**.** A signal $f : S^2 \to \mathbb{R}$ is *band-limited* at degree L if its spherical harmonic expansion

$$f(\theta, \varphi) = \sum_{\ell=0}^{L} \sum_{m=-\ell}^{\ell} a_\ell^m \, Y_\ell^m(\theta, \varphi) \tag{5}$$

has $a_\ell^m = 0$ for all $\ell > L$. We write $\mathbf{F}_\ell = (a_\ell^{-\ell}, \ldots, a_\ell^\ell) \in \mathbb{C}^{2\ell+1}$ for the coefficient vector at degree ℓ.

For real f, the conjugation symmetry $a_\ell^{-m} = (-1)^m \overline{a_\ell^m}$ holds, so $\mathbf{F}_\ell$ has $2\ell + 1$ real degrees of freedom. The total signal dimension is $\dim = \sum_{\ell=0}^{L}(2\ell + 1) = (L + 1)^2$.

Definition 4 (Rotation action)**.** Under $g \in \mathrm{SO}(3)$, the spherical harmonic coefficients transform as $\mathbf{F}_\ell \mapsto D^\ell(g)\,\mathbf{F}_\ell$, where D^ℓ is the $(2\ell + 1)$-dimensional Wigner D-matrix.

In the irrep notation of Mataigne et al. [2024], $\{\rho_\ell\}_{\ell\geq 0}$ are the (unitary) irreducible representations of $\mathrm{SO}(3)$ with V_{ρ_ℓ} the $(2\ell+1)$-dimensional space of degree-ℓ spherical harmonics and $\rho_\ell(g) = D^\ell(g)$; the Fourier coefficient of f at ρ_ℓ is $\mathcal{F}(f)_{\rho_\ell} = \mathbf{F}_\ell$. We use $\mathbf{F}_\ell$ as the working symbol throughout.

Definition 5 (Bispectrum on S^2)**.** For each triple (ℓ_1, ℓ_2, ℓ) satisfying the triangle inequality $|\ell_1 - \ell_2| \leq \ell \leq \ell_1 + \ell_2$, the *bispectral coefficient* is

$$\beta_{\ell_1,\ell_2,\ell} = \sum_{\substack{m_1,\, m_2,\, m \\ m_1+m_2=m}} C^{\ell,\, m}_{\ell_1,\, m_1;\, \ell_2,\, m_2}\; a_{\ell_1}^{m_1}\, a_{\ell_2}^{m_2}\, \overline{a_\ell^m}, \tag{6}$$

where $C^{\ell,\, m}_{\ell_1,\, m_1;\, \ell_2,\, m_2}$ denotes the Clebsch–Gordan coefficient $\langle \ell_1, m_1;\, \ell_2, m_2 \mid \ell, m\rangle$. Each $\beta_{\ell_1,\ell_2,\ell}$ is a scalar; for real f it is real when $\ell_1+\ell_2+\ell$ is even and purely imaginary when $\ell_1+\ell_2+\ell$ is odd (Lemma 28).

Definition 5 is the specialization of the general G-bispectrum (eq. (2)) to the case $G = \mathrm{SO}(3)$ acting on S^2, where the irreps are Wigner-D matrices and the Fourier coefficients are spherical harmonic coefficients a_ℓ^m.

Definition 6 (CG power spectrum, optional diagnostic)**.** For a triple (ℓ_1, ℓ_2, ℓ) satisfying the triangle inequality, write $(\mathbf{F}_{\ell_1} \otimes \mathbf{F}_{\ell_2})\big|_\ell$ for the projection of the Kronecker product $\mathbf{F}_{\ell_1} \otimes \mathbf{F}_{\ell_2}$ onto the D^ℓ-isotypic component, realised by contracting with the CG coefficients $C^{\ell,\, m}_{\ell_1,\, m_1;\, \ell_2,\, m_2}$. The *CG power spectrum entry* is the squared norm of this projection,

$$P_{\ell_1,\ell_2,\ell} := \|(\mathbf{F}_{\ell_1} \otimes \mathbf{F}_{\ell_2})|_\ell\|^2 = \sum_{m=-\ell}^{\ell} \Bigg| \sum_{\substack{m_1,m_2 \\ m_1+m_2=m}} C^{\ell,\, m}_{\ell_1,\, m_1;\, \ell_2,\, m_2}\; a_{\ell_1}^{m_1}\, a_{\ell_2}^{m_2} \Bigg|^2. \tag{7}$$

This is a degree-4, real-valued, $\mathrm{SO}(3)$-invariant polynomial in the signal coefficients. CG power entries are an *optional* implementation diagnostic and are not used in the conditional completeness theorem; we set $\mathcal{S}_P = \varnothing$ throughout the formal proof. The library may append a $O(L^2)$ family of CG power scalars without affecting completeness or the asymptotic output size.

Proposition 7 (Size of the full bispectrum)**.** *The number of admissible triples* (ℓ_1, ℓ_2, ℓ) *with* $0 \leq \ell_1, \ell_2, \ell \leq L$ *satisfying the triangle inequality* $|\ell_1 - \ell_2| \leq \ell \leq \ell_1 + \ell_2$ *is* $\Theta(L^3)$.

Proof. For fixed ℓ_1 and ℓ_2, the triangle inequality admits $\ell \in \{|\ell_1 - \ell_2|, \ldots, \min(\ell_1 + \ell_2, L)\}$, giving $\min(\ell_1 + \ell_2, L) - |\ell_1 - \ell_2| + 1$ valid values of ℓ. Summing:

$$N_{\text{full}} = \sum_{\ell_1=0}^{L} \sum_{\ell_2=0}^{L} \big(\min(\ell_1 + \ell_2, L) - |\ell_1 - \ell_2| + 1\big) = \tfrac{1}{3}L^3 + \mathcal{O}(L^2).$$

□

Remark 8 (Invariance). Each $\beta_{\ell_1,\ell_2,\ell}$ is $\mathrm{SO}(3)$-invariant: the Wigner matrices satisfy $D^{\ell_1} \otimes D^{\ell_2} = \bigoplus_\ell D^\ell$ under the CG decomposition, so the triple product contracted with CG coefficients is unchanged by $\mathbf{F}_\ell \mapsto D^\ell(g)\mathbf{F}_\ell$.

A.2 Why $\mathcal{O}(L)$ is impossible

Proposition 9 (Dimensional lower bound)**.** *Any* smooth or polynomial *complete* $\mathrm{SO}(3)$*-invariant for band-limited signals on* S^2 *at band-limit* L *must have codomain of dimension at least* $(L+1)^2-3$.

Proof. Let $\Phi : \mathbb{R}^{(L+1)^2} \to \mathbb{R}^k$ be a smooth (or polynomial) $\mathrm{SO}(3)$-invariant map that separates orbits on a Zariski-open dense subset $U \subset \mathbb{R}^{(L+1)^2}$. Smoothness implies Φ has a well-defined generic rank $r := \max_{x\in U} \operatorname{rank} d\Phi_x$; on the open dense subset where this rank is attained, Φ is locally a submersion onto an r-dimensional submanifold, and orbit-separation forces $r \geq \dim(\mathbb{R}^{(L+1)^2}/\mathrm{SO}(3)) = (L+1)^2 - \dim \mathrm{SO}(3) = (L+1)^2 - 3$ (generic-rank theorem for $\mathrm{SO}(3)$ acting on V_L; cf. Tarski–Seidenberg in the polynomial case). Hence $k \geq r \geq (L+1)^2 - 3 = \Omega(L^2)$. □

Remark 10. The chain $\beta_{1,\ell,\ell+1}$ for $\ell = 0, \ldots, L-1$ provides only $\sim 3L$ scalar constraints (since each chain entry couples $\mathbf{F}_1 \in \mathbb{C}^3$ with one known $\mathbf{F}_\ell$). This cannot recover $(L+1)^2 \approx L^2$ unknowns.

Remark 11 (Smoothness assumption). The smoothness/polynomial qualifier is essential: pathological non-measurable invariants (e.g. a fixed orbit-section indexed by the axiom of choice) can in principle have codomain dimension 1 but are not implementable. All practical bispectral or spectral invariants are polynomial, so the proposition applies.

A.3 The selective index set

The goal is to select, for each target degree $\ell \geq 8$, exactly $2\ell+1$ bispectral triples (ℓ_1, ℓ_2, ℓ) such that the selected entries are simultaneously linear in $\mathbf{F}_\ell$ (or in $\overline{\mathbf{F}_\ell}$, which we conjugate) and have a coefficient matrix depending only on already-recovered lower degrees. Combined with a constant-size full-bispectrum seed at degrees $\leq L_{\mathrm{seed}} = \min(L, 4)$, this gives an $O(L^2)$-sized invariant matching the orbit-space lower bound of Proposition 9.

Definition 12 (Selective $\mathrm{SO}(3)$ bispectrum on S^2)**.** Set $L_{\mathrm{seed}} := \min(L, 4)$. The *seed index set* is the full bispectrum truncated at L_{seed},

$$\mathcal{S}_{\mathrm{seed}} := \{(\ell_1, \ell_2, \ell) : 0 \leq \ell_1 \leq \ell_2 \leq L_{\mathrm{seed}},\ |\ell_1 - \ell_2| \leq \ell \leq \min(\ell_1 + \ell_2, L_{\mathrm{seed}})\},$$

of size at most $(L_{\mathrm{seed}}+1)^3 \leq 125$, independent of L once $L \geq 4$. This matches the implementation seed $L_0 = 4$ of Algorithm 2, on which Bandeira et al.'s exact-arithmetic Gröbner-basis evidence for the scalar bispectrum at low degrees [Bandeira et al., 2017] is strongest. For each intermediate target $5 \leq \ell \leq 7$ we use the small explicit linear block $\mathcal{T}_\ell^{\mathrm{small}}$ tabulated in Algorithm 2; for each high-degree target $\ell \geq 8$ the *bootstrap block* $\mathcal{T}_\ell$ (also used in Proposition 13) is

$$\begin{aligned} \mathcal{T}_\ell := &\{(a, \ell, \ell - a) : 1 \leq a \leq \ell - 1\} \\ &\cup \big\{(a, \ell, \ell - a + 1) : 2 \leq a \leq \ell - 1,\ a \neq (\ell+1)/2\big\} \\ &\cup \{(a, \ell - a, \ell) : 1 \leq a \leq 4\} \ \cup\ \mathcal{Z}_\ell, \end{aligned} \tag{8}$$

where $\mathcal{Z}_\ell = \emptyset$ for ℓ even and $\mathcal{Z}_\ell = \{(2, \ell-1, \ell)\}$ for ℓ odd ($\ell \geq 9$). The exclusion $a \neq (\ell+1)/2$ for odd ℓ removes the self-conjugate triple $(r, 2r-1, r)$ at $\ell = 2r-1$, which has a repeated index and is identically zero on real signals (Proposition 30); the compensating row $(2, \ell-1, \ell) \in \mathcal{Z}_\ell$ is admissible ($|\ell - 1 - 2| \leq \ell \leq \ell - 1 + 2$), has all-distinct indices, and is not contained in the chain family above (which only covers $(a, \ell-a, \ell)$ with $a = 1, \ldots, 4$, i.e. pairs $(a, \ell-a)$ with $a \leq 4$). The size counts to $(\ell-1) + (\ell-2) + 4 = 2\ell+1$ for even ℓ and $(\ell-1) + (\ell-3) + 4 + 1 = 2\ell+1$ for odd $\ell \geq 9$. The bispectral index set is

$$\mathcal{S}_\beta := \mathcal{S}_{\mathrm{seed}} \ \cup \bigcup_{\ell=5}^{L} \mathcal{T}_\ell,$$

where $\mathcal{T}_\ell = \mathcal{T}_\ell^{\mathrm{small}}$ for $5 \leq \ell \leq 7$ (Algorithm 2) and is given by eq. (8) for $\ell \geq 8$. The *selective bispectrum* $\beta_{\mathrm{sel}} : V_L \to \mathbb{R}^N$ is the concatenation of all selected scalar entries $\{\beta_{\ell_1,\ell_2,\ell}\}_{(\ell_1,\ell_2,\ell)\in\mathcal{S}_\beta}$; by Lemma 28, each entry contributes one real component on the real-signal slice (real for even parity, purely imaginary for odd parity). We set $\mathcal{S}_P := \varnothing$ in the formal theorem; the optional CG power augmentation (Definition 6) may be appended at $O(L^2)$ cost without changing the asymptotic budget or the conclusion of Theorem 18.

Library implementation note. Algorithm 2 below specifies the implementation-grade index construction used by the `SO3onS2` module. It augments Definition 12 in two ways: (i) it includes *mandatory even self-couplings* $\beta_{\ell,\ell,\ell'}$ (ℓ' even, $2 \leq \ell' \leq \ell$) and the power entries $\beta_{0,\ell,\ell}$ for numerical robustness on the real-signal slice, and (ii) it greedily appends CG-power entries to $\mathcal{S}_P$ to

maximise the per-degree real Jacobian rank. The augmented invariant is a strict superset of β_{sel}, so completeness transfers automatically; the formal theorem is stated for the smaller, purely-bispectral β_{sel}.

Proposition 13 (Bootstrap-block entry count)**.** *For each target degree $\ell \geq 8$, the bootstrap block $\mathcal{T}_\ell$ defined in* (8) *contains exactly $2\ell{+}1$ admissible triples and they are pairwise distinct. Each triple in $\mathcal{T}_\ell$ has ℓ in exactly one of the three slots, with the other two slots filled by indices strictly smaller than ℓ, so each entry contributes one scalar equation linear in either $a_\ell^{m_2}$ or $\overline{a_\ell^m}$ with coefficients depending only on lower-degree $\mathbf{F}_{\ell'}$ ($\ell' < \ell$).*

Proof. The three subfamilies in (8) have sizes $\ell{-}1$, $\ell{-}2$, 4, summing to $2\ell{+}1$. Triangle-inequality and bound checks ($a \leq \ell{-}1$, $\ell{-}a \geq 1$, $\ell{-}a{+}1 \leq \ell$) show every triple is admissible. Pairwise distinctness is immediate from the disjoint patterns of the position of the index ℓ. Linearity in the ℓ-slot: the first two families place ℓ in the ℓ_2-slot of (6), so $a_\ell^{m_2}$ enters once with known $a_a^{m_1}$ and $\overline{a_{\ell-a}^m}$ (resp. $\overline{a_{\ell-a+1}^m}$); the third family places ℓ in the ℓ-slot, so $\overline{a_\ell^m}$ enters once with known $a_a^{m_1}$ and $a_{\ell-a}^{m_2}$. □

Proposition 14 (Output size)**.** *For every $L \geq 4$,*

$$|\beta_{\mathrm{sel}}| \;=\; |\mathcal{S}_{\mathrm{seed}}| + \sum_{\ell=5}^{L}(2\ell{+}1) \;=\; \Theta(L^2),$$

counting one real component per scalar entry (real for even parity, purely imaginary for odd parity, by Lemma 28). This matches the orbit-space lower bound $(L{+}1)^2 - 3$ of Proposition 9 up to an additive constant (no factor-of-two from complex storage). For $L = 4$ the sum is empty and $|\beta_{\mathrm{sel}}| = |\mathcal{S}_{\mathrm{seed}}| = O(1)$. The augmented Jacobian-rank diagnostic (Remark 38) is reported only for the live, non-vanishing entries; the count matches the 30 *(resp.* 49*) live entries reported by the implementation at $L = 4$ (resp. $L = 5$).*

A.4 Completeness proof

The proof proceeds in four stages: (1) closed-form recovery of A_0, $\mathbf{F}_1$ and $\mathrm{SO}(3)$ gauge fixing (Lemma 19, Lemma 21); (2) seed completeness for degrees $\leq L_{\mathrm{seed}}$ from the full-bispectrum seed block $\mathcal{S}_{\mathrm{seed}}$ (Hypothesis 25, motivated by Kakarala [Kakarala, 1992, 2012] and Bandeira et al. [Bandeira et al., 2017]); (3) odd-parity T_R-resolution from the seed (Lemma 28, Corollary 32); (4) per-degree linear bootstrap for $\ell \geq 8$ via the bootstrap block $\mathcal{T}_\ell$ (Hypothesis 34). Stages (2) and (4) are the two named hypotheses H1 and H2 of Section 6; the remaining stages are unconditional. The constructive seed solver of Appendix B is an alternative, well-conditioned algorithmic route to stage (2) but is not load-bearing for the conditional theorem.

Definition 15 (Reflection map T_R)**.** The azimuthal reflection $R : (\theta, \varphi) \mapsto (\theta, -\varphi)$ acts on spherical harmonic coefficients as $a_\ell^m \mapsto (-1)^m a_\ell^{-m}$. For real signals, this equals complex conjugation: $a_\ell^m \mapsto \overline{a_\ell^m}$. Since $R \in \mathrm{O}(3) \setminus \mathrm{SO}(3)$, the signal $R \cdot f$ is generically in a *different* $\mathrm{SO}(3)$-orbit from f.

Assumption 16 (Genericity)**.** The signal $f \in V_L$ satisfies:

- (G1) $a_0^0 \neq 0$ (non-zero mean).
- (G2) $\|\mathbf{F}_1\| \neq 0$ (non-zero dipole).
- (G3) After gauge-fixing (Lemma 19), $a_2^1 \neq 0$ (azimuthal gauge is resolvable).
- (G4) $\mathrm{Im}(\beta_{2,3,4}(f)) \neq 0$ (the first odd-parity entry of the seed block is non-vanishing, so $T_R \cdot f$ is detected; cf. Corollary 32).
- (G5) For every $\ell \geq 8$, the bootstrap coefficient matrix $A_\ell(\mathbf{F}_{<\ell})$ assembled from $\{\beta_{\ell_1,\ell_2,\ell}\}_{(\ell_1,\ell_2,\ell)\in\mathcal{T}_\ell}$ has full real rank $2\ell{+}1$ (the pointwise version of Hypothesis 34).

Remark 17. Each of (G1)–(G4) excludes a proper algebraic subvariety of $\mathbb{R}^{(L+1)^2}$, so their union is Lebesgue-null. Condition (G5) is the pointwise instance of Hypothesis H2 evaluated at f: under Hypothesis 34 the failure set of (G5) is contained in the union of polynomial varieties $\{\Delta_\ell = 0\}_{\ell=8}^{L}$, also a proper algebraic subvariety.

Theorem 18 (Conditional completeness of β_{sel})**.** *Let $L \geq 4$ and assume Hypothesis H1 (Hypothesis 25, finite seed injectivity at $L_{\mathrm{seed}} = \min(L, 7)$) and, if $L \geq 8$, Hypothesis H2 (Hypothesis 34,*

uniform bootstrap-determinant nonvanishing). For any $f, f' \in V_L$ satisfying Assumption 16, if $\beta_{\mathrm{sel}}(f) = \beta_{\mathrm{sel}}(f')$, then $f' = g \cdot f$ for some $g \in \mathrm{SO}(3)$. The generic set on which the conclusion holds is the complement of a proper algebraic subvariety of V_L, in particular of full Lebesgue measure. The output β_{sel} has $\Theta(L^2)$ real components, matching the orbit-space lower bound $(L+1)^2-3$ of Proposition 9 up to a constant factor.

Lemma 19 (Gauge fixing)**.** *For generic $f \in V_L$ (with $\|\mathbf{F}_1\| \neq 0$ and $a_2^1 \neq 0$ after aligning $\mathbf{F}_1$), there exists a unique $g \in \mathrm{SO}(3)$ such that the gauge-fixed signal $\tilde{f} = g \cdot f$ satisfies:*

(i) $a_1^{-1} = a_1^1 = 0, \quad a_1^0 = c > 0$ *(alignment of $\mathbf{F}_1$ to the z-axis),*
(ii) $\mathrm{Im}(a_2^1) = 0, \quad a_2^1 = x > 0$ *(fixing the residual $\mathrm{SO}(2)_z$ symmetry).*

This consumes all 3 degrees of freedom of $\mathrm{SO}(3)$.

Proof. The $\ell = 1$ Wigner D-matrix $D^1(g)$ is the standard 3×3 rotation matrix. Since $\mathrm{SO}(3)$ acts transitively on S^2, any nonzero $\mathbf{F}_1$ can be rotated to align with $\hat{z}$, leaving an $\mathrm{SO}(2)_z$ stabiliser. Choosing $\varphi = -\arg(a_2^1)$ yields $a_2^1 \in \mathbb{R}_{>0}$, exhausting all three rotational degrees of freedom; equivalently the gauge-fixed azimuthal angle is unique in $[0, 2\pi)$. Without gauge fixing the $\mathrm{SO}(2)_z$ stabiliser leaves a one-parameter family of equivalent solutions, and any further $C_n \subset \mathrm{SO}(2)_z$ subgroup symmetry would discretise this family to n representatives. □

Remark 20 (Gauge-fixed parameterisation). On the gauge slice, the real parameters are:

$$A_0 = a_0^0, \quad c = a_1^0, \quad (y, x, u, v) = (a_2^0, \mathrm{Re}\, a_2^1, \mathrm{Re}\, a_2^2, \mathrm{Im}\, a_2^2),$$

and for $\ell \geq 3$: $t_\ell = a_\ell^0$, and $(p_{\ell,m}, q_{\ell,m}) = (\mathrm{Re}\, a_\ell^m, \mathrm{Im}\, a_\ell^m)$ for $m = 1, \ldots, \ell$. The reflection T_R (Definition 15) acts on the gauge slice as:

$$T_R: \quad v \mapsto -v, \qquad q_{\ell,m} \mapsto -q_{\ell,m} \quad \forall\, \ell \geq 3,\ m \geq 1. \tag{9}$$

All other gauge-fixed parameters are unchanged. T_R preserves the gauge conditions (i)–(ii).

Lemma 21. *For $a_0^0 \neq 0$: $\beta_{0,0,0} = (a_0^0)^3$ determines $A_0 = a_0^0$ uniquely. Then $\beta_{0,1,1} = A_0 \|\mathbf{F}_1\|^2$ determines $c = \|\mathbf{F}_1\|$, and $\mathbf{F}_1 = (0, c, 0)^\top$ is fixed by the gauge.*

A.4.1 Structural seed completeness via Kakarala's theorem

Scalar versus matrix bispectrum. Kakarala's Theorem 7 [Kakarala, 2012, 1992] is stated for the *matrix* (or *operator*) bispectrum: each Fourier mode is a $(2\ell+1) \times (2\ell_1+1)(2\ell_2+1)$ operator, and the triple correlation contains every spin component of $\mathbf{F}_\ell$ before any CG contraction. Our scalar bispectrum $\beta_{\ell_1,\ell_2,\ell}$ (Definition 5) is the single Clebsch–Gordan contraction of that operator down to one complex number per triple. The contraction discards the $(2\ell+1)$-dimensional vector index that Kakarala's argument relies on, so Theorem 7 does *not* directly imply injectivity of the scalar map. We retain Kakarala's theorem below (Lemma 22) as *structural motivation*: it shows that a G/H-bispectrum carries enough information in the matrix sense, and a sufficiently rich finite collection of scalar contractions (the seed) is conjectured to inherit the same separation power on a Zariski-open set of signals. This conjecture is exactly hypothesis 25 (H1). Bandeira et al. [Bandeira et al., 2017] provide an independent Gröbner-basis check for the scalar setup at low band-limits, but no closed-form proof is known beyond their finite-degree range.

Lemma 22 (Kakarala bispectrum completeness on S^2, restated)**.** *Let $G = \mathrm{SO}(3)$ and $H = \mathrm{SO}(2)_z \subset G$ be the stabiliser of the north pole, so that $G/H \cong S^2$. Let $r, s \in L^1(G)$ be left H-invariant functions whose Fourier coefficients $R(\ell), S(\ell)$ all satisfy the maximal H-rank condition (equivalently, for S^2 functions, the spherical-harmonic vectors $\mathbf{F}_\ell$ are nonzero for every ℓ). Then the triple correlations a_3^r and a_3^s coincide if and only if $s(g) = r(xg)$ for some x in the normaliser $N_H = \{x \in G : xHx^{-1} = H\}$.*

Proof. This is [Kakarala, 2012, Theorem 7], originally proved in [Kakarala, 1992]; the proof proceeds via the Iwahori–Sugiura duality theorem for homogeneous spaces of compact groups. An independent computer-algebra certificate of the same conclusion for $L \leq 15$ (Gröbner-basis elimination, no numerical threshold) is given by Bandeira et al. [2017], which we invoke below as a backup whenever a structural reader prefers to avoid relying on the thesis [Kakarala, 1992] for the proof of Theorem 7. □

Remark 23 (Computation of N_H and absence of an $\mathrm{O}(3)$ ambiguity). For $H = \mathrm{SO}(2)_z = \{R_z(\varphi) : \varphi \in \mathbb{R}\}$, an element $x \in \mathrm{SO}(3)$ lies in N_H iff $x \cdot \hat{z} = \pm\hat{z}$, hence $N_H = H \cup R_x(\pi) \cdot H$, the dihedral group $\mathrm{O}(2)$ realised *inside* $\mathrm{SO}(3)$. In particular, $N_H \subset \mathrm{SO}(3)$: the non-trivial coset representative $R_x(\pi)$ is a proper $\mathrm{SO}(3)$-rotation (an azimuthal-axis flip), *not* an $\mathrm{O}(3) \setminus \mathrm{SO}(3)$ reflection. Therefore Lemma 22 immediately implies $\mathrm{SO}(3)$-orbit recovery on S^2: equality of bispectra forces s and r to lie in the same $\mathrm{SO}(3)$-orbit, and the residual $N_H/H = \mathbb{Z}/2$ is internal to $\mathrm{SO}(3)$ (it merely reflects that two distinct H-invariant lifts of the same S^2-function exist along an $\mathrm{SO}(3)$-orbit). The orientation-reversing reflection $T_R : a_\ell^m \mapsto (-1)^m \overline{a_\ell^m}$ (Definition 15) is a separate operation: $T_R \in \mathrm{O}(3) \setminus \mathrm{SO}(3)$ flips the sign of every odd-parity bispectral entry (Lemma 28), so $T_R \cdot f$ is generically detected as a different signal by the full bispectrum.

Definition 24 (Seed block of β_{sel})**.** We use the seed index set $\mathcal{S}_{\mathrm{seed}}$ of Definition 12, that is, the full bispectrum at $L_{\mathrm{seed}} = \min(L, 4)$. The number of seed entries is $N_{\mathrm{seed}} \leq (L_{\mathrm{seed}}+1)^3 \leq 125$, *constant* once $L \geq 4$. The output of β_{sel} restricted to the seed is therefore $O(1)$ scalars, and the total $|\beta_{\mathrm{sel}}|$ remains $\Theta(L^2)$ (Proposition 14).

Hypothesis 25 (H1: finite seed injectivity)**.** Let $L_{\mathrm{seed}} = \min(L, 4)$. For every $f \in V_{L_{\mathrm{seed}}}$ satisfying Assumption 16 (in particular $\mathbf{F}_\ell \neq 0$ for $0 \leq \ell \leq L_{\mathrm{seed}}$), the full *scalar* bispectrum on $\mathcal{S}_{\mathrm{seed}}$ together with the $\mathrm{O}(3)$-distinguishing entry $\mathrm{Im}(\beta_{2,3,4})$ uniquely determines $(\mathbf{F}_0, \mathbf{F}_1, \ldots, \mathbf{F}_{L_{\mathrm{seed}}})$ on the gauge slice of Lemma 19.

Structural support for H1. Two independent partial results motivate H1, but neither alone is a proof of the band-limited *scalar* statement above (recall the matrix-vs-scalar gap of Paragraph A.4.1):

(i) *Representation-theoretic (Kakarala).* Theorem 7 of [Kakarala, 2012], restated as Lemma 22 and originally proved in [Kakarala, 1992], gives recovery up to the normaliser $N_H \subset \mathrm{SO}(3)$ from the *matrix-valued* bispectrum on $G = \mathrm{SO}(3)$, $H = \mathrm{SO}(2)_z$, in the $L^1(G)$ category; Remark 23 computes N_H explicitly. Theorem 7 does not directly imply scalar-bispectrum injectivity, and the translation to a band-limited *scalar* statement at $L_{\mathrm{seed}} = 4$ is not formalised in either [Kakarala, 1992, 2012]; H1 is the band-limited scalar specialisation we assume.

(ii) *Computer-algebra (Bandeira et al.).* Bandeira et al. [2017] verify by symbolic computation (Jacobian rank of the scalar bispectrum map at a rational test point, combined with Zariski-density propagation) that the scalar bispectrum separates generic $\mathrm{SO}(3)$-orbits on $V_0 \oplus \cdots \oplus V_{L_0}$ for every $L_0 \leq 15$. This is a one-time computer-algebra certificate over an algebraic number field rather than a floating-point witness, and directly covers our fixed $L_{\mathrm{seed}} = 4$.

The constructive seed solver of Appendix B is a third, explicitly algorithmic route to the seed (closed-form $\ell \leq 2$, finite-fibre $\ell = 3$, degree-4 filter, parity-based T_R resolution); it is an alternative to assuming H1 inside the implementation, but does not reduce H1 to a formally proved statement in the band-limited setting.

Remark 26 (On the "up to reflection" caveat of Edidin and Satriano [2024]). The reflection caveat reduces to three points:

(1) $T_R : a_\ell^m \mapsto (-1)^m \overline{a_\ell^m}$ lies in $\mathrm{O}(3) \setminus \mathrm{SO}(3)$, not in $\mathrm{SO}(3)$ (so T_R is not absorbed by the gauge of Lemma 19).
(2) Every odd-parity scalar entry flips sign under T_R by Lemma 28.
(3) $\mathrm{Im}(\beta_{2,3,4})$ is generically nonzero (Cor. 32), so its sign separates f from $T_R \cdot f$ at any $L \geq 4$.

The reflection caveat in Edidin and Satriano [2024, Section 1] refers to the constructive polar-decomposition algorithm of Kakarala [Kakarala, 2012, Section 5], not to the bispectrum data itself; Theorem 7 there gives recovery up to $N_H \subset \mathrm{SO}(3)$ and the odd-parity entry above closes the remaining T_R ambiguity.

Remark 27 (Relation to the constructive seed lemmas). Appendix B provides an *explicit constructive solver* for the seed: closed-form recovery at degree 2, finite-fibre enumeration at degree 3, degree-4 filtering, and an explicit parity-based T_R resolution. This algorithmic content is used by the implementation. It is an alternative algorithmic route to the seed step (Hypothesis 25, H1) but is not load-bearing for the conditional completeness theorem, which assumes H1 directly. The constructive solver remains useful because it reduces the seed to a low-dimensional, well-conditioned numerical solve.

A.4.2 Parity structure of the bispectrum

Lemma 28 (Parity of the bispectrum)**.** *For any signal (real or complex):*

$$\beta_{\ell_1,\ell_2,\ell}(T_R \cdot f) = (-1)^{\ell_1+\ell_2+\ell}\, \beta_{\ell_1,\ell_2,\ell}(f). \tag{10}$$

For real *signals, there is an independent constraint:*

$$\overline{\beta_{\ell_1,\ell_2,\ell}(f)} = (-1)^{\ell_1+\ell_2+\ell}\, \beta_{\ell_1,\ell_2,\ell}(f). \tag{11}$$

Together:

(a) If $\ell_1+\ell_2+\ell$ is even: β is real-valued (by (11)*) and T_R-invariant (by* (10)*).*
(b) If $\ell_1+\ell_2+\ell$ is odd: β is purely imaginary (by (11)*) and $\beta(T_R \cdot f) = -\beta(f)$, so $\operatorname{Im}\beta$ flips sign under T_R.*

Proof. Under T_R, each $a_\ell^m \mapsto (-1)^m a_\ell^{-m}$. Substituting into $\beta = \sum_m (\sum_{m_1+m_2=m} C\, a^{m_1} a^{m_2})\overline{a^m}$, the three sign factors $(-1)^{m_1+m_2+m} = (-1)^{2m} = 1$ cancel. Re-indexing $m \mapsto -m$ and applying the CG symmetry $C^{\ell,\,-m}_{\ell_1,\,-m_1;\,\ell_2,\,-m_2} = (-1)^{\ell_1+\ell_2-\ell}\, C^{\ell,\,m}_{\ell_1,\,m_1;\,\ell_2,\,m_2}$ yields (10) (with $(-1)^{\ell_1+\ell_2-\ell} = (-1)^{\ell_1+\ell_2+\ell}$).

For real signals, $\overline{a_\ell^m} = (-1)^m a_\ell^{-m}$. Taking the complex conjugate of β and re-indexing similarly yields (11). □

Lemma 29 (CG power is T_R-invariant)**.** *For all triples (ℓ_1,ℓ_2,ℓ): $P_{\ell_1,\ell_2,\ell}(T_R \cdot f) = P_{\ell_1,\ell_2,\ell}(f)$.*

Proof. Under T_R, the CG-projected component transforms as $(\mathbf{F}_{\ell_1} \otimes \mathbf{F}_{\ell_2})|_\ell^M \mapsto (-1)^{\ell_1+\ell_2+\ell+M}\, (\mathbf{F}_{\ell_1} \otimes \mathbf{F}_{\ell_2})|_\ell^{-M}$. Taking the squared norm and re-indexing the sum over M leaves P unchanged. □

Proposition 30 (Vanishing of odd-parity entries with a repeated index)**.** *Let $\beta_{\ell_1,\ell_2,\ell}$ be the scalar bispectrum entry of a real signal f (so $\ell_1+\ell_2+\ell$ odd). If the index multiset $\{\ell_1,\ell_2,\ell\}$ contains a repeated value, then $\beta_{\ell_1,\ell_2,\ell}(f) \equiv 0$ as a polynomial in the SH coefficients of f.*

Proof. Define the symmetric scalar 3j contraction

$$T_{\ell_1,\ell_2,\ell_3}(f) := \sum_{m_1,m_2,m_3} \begin{pmatrix} \ell_1 & \ell_2 & \ell_3 \\ m_1 & m_2 & m_3 \end{pmatrix} a_{\ell_1}^{m_1}\, a_{\ell_2}^{m_2}\, a_{\ell_3}^{m_3}.$$

Converting the Wigner 3j into a Clebsch-Gordan coefficient via $\left(\begin{smallmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{smallmatrix}\right) = \frac{(-1)^{j_1-j_2-m_3}}{\sqrt{2j_3+1}}\, C^{j_3,\,-m_3}_{j_1,\,m_1;\,j_2,\,m_2}$ and applying the reality identity $a_{\ell_3}^{m_3} = (-1)^{m_3}\, \overline{a_{\ell_3}^{-m_3}}$ shows

$$\beta_{\ell_1,\ell_2,\ell}(f) = (-1)^{\ell_1+\ell_2}\sqrt{2\ell+1}\, T_{\ell_1,\ell_2,\ell}(f),$$

so β and T have the same vanishing locus. The 3j symbol is invariant under even column permutations and picks up $(-1)^{\ell_1+\ell_2+\ell_3}$ under odd ones. If two columns i,j satisfy $\ell_i = \ell_j$, the inserted vectors a_{ℓ_i}, a_{ℓ_j} are literally the same coefficient vector of f, so swapping those two columns leaves the contracted sum T unchanged while multiplying the 3j prefactor by $(-1)^{\ell_1+\ell_2+\ell_3}$. For odd parity this forces $T = -T$, hence $T \equiv 0$ and $\beta \equiv 0$. □

Remark 31 (Consequences for $\mathcal{T}_\ell$). Proposition 30 applies in particular to the family $(r, 2r-1, r)$: it has parity $4r-1$ (odd) and a repeated index, so $\beta_{r,2r-1,r} \equiv 0$ on real signals. Numerical evaluation at the seed-42 real witness confirms this: for $r = 2,\ldots,7$ the entry is zero to double-precision rounding ($|\beta| \lesssim 10^{-14}$). Any formal definition of $\mathcal{T}_\ell$ must therefore exclude these triples (see (8)); we substitute a chain row with all-distinct indices in their place.

Corollary 32 (First nonzero odd-parity entry)**.** *The first bispectral entry with odd parity and generically nonzero imaginary part occurs at degree $\ell = 4$: $\beta_{2,3,4}$, with $\ell_1+\ell_2+\ell = 9$ (odd) and ℓ_1,ℓ_2,ℓ all distinct. At the deterministic witness (seed 42), $\operatorname{Im}(\beta_{2,3,4}) \approx 3.26 \neq 0$, and $\operatorname{Im}(\beta_{2,3,4})(T_R \cdot f) \approx -3.26$. Similarly, $\beta_{2,4,3}$ and $\beta_{3,4,2}$ are nonzero odd-parity entries at degree 4.*

Remark 33 (No parity-breaking at $\ell \le 3$). For degrees ≤ 3, every odd-parity triple (ℓ_1,ℓ_2,ℓ) has at least two indices equal (since $\max(\ell_1,\ell_2,\ell) \le 3$ and $\ell_1+\ell_2+\ell$ odd forces at least one pair to be equal in this range). By Proposition 30, all odd-parity entries vanish. Therefore, the T_R ambiguity cannot be resolved using only degrees ≤ 3.

A.4.3 Per-degree recovery ($\ell \geq 5$)

Hypothesis 34 (H2: uniform per-degree augmented Jacobian nonvanishing)**.** For each $\ell \geq 5$, let $J_\ell(\mathbf{F}_{<\ell})$ be the real Jacobian of the augmented per-degree block

$$\beta_\ell^{\mathrm{aug}}(f) := \{\beta_{\ell_1,\ell_2,\ell}(f) : (\ell_1,\ell_2,\ell) \in \mathcal{T}_\ell \cup \mathcal{S}_\ell^{\mathrm{self}} \cup \mathcal{S}_\ell^P\},$$

with respect to the $2\ell{+}1$ real parameters of $\mathbf{F}_\ell$ on the real-signal slice $\overline{a_{\ell'}^m} = (-1)^m a_{\ell'}^{-m}$ for all $\ell' \leq \ell$. Here $\mathcal{S}_\ell^{\mathrm{self}}$ is the set of mandatory even self-couplings $\{(\ell,\ell,\ell') : 0 \leq \ell' \leq \ell,\ \ell' \text{ even}\}$ and $\mathcal{S}_\ell^P$ the set of CG-power entries at target degree ℓ generated by Algorithm 2. Then J_ℓ has full column rank $2\ell{+}1$ on a Zariski-open subset of the real-signal lower-degree coordinate space.

Remark 35 (Why the augmented block is needed). The bootstrap block $\mathcal{T}_\ell$ alone is rank-deficient on the real-signal slice: at the seed-42 real witness, the $(2\ell{+}1) \times (2\ell{+}1)$ submatrix A_ℓ assembled from the $\mathcal{T}_\ell$ rows has SVD rank $\approx \ell{+}2$ rather than $2\ell{+}1$ for $\ell \in \{8,\ldots,15\}$ (verified numerically; the deficit is generic over real witnesses). The mandatory even self-couplings and CG-power scalars are quadratic or cubic in $\mathbf{F}_\ell$ and supply the additional independent real equations needed to fill out the per-degree real Jacobian. H2 is therefore stated for the augmented block, not for A_ℓ alone.

Numerical evidence for H2. Per-degree augmented Jacobians J_ℓ are evaluated at the deterministic seed-42 real-signal witness in IEEE 754 double precision (scripts: `benchmarks/verify_linear_bootstrap.py`, `paper/exact_certificates.py`). For every $\ell \in \{5,\ldots,100\}$ the smallest singular value satisfies $\sigma_{\min}(J_\ell)/\sigma_{\max}(J_\ell) \geq 10^{-4}$, well above the SVD numerical noise floor at this size. This is consistent with H2 being non-degenerate at the witness, but is a numerical *witness* at one rational floating-point point per degree, not a closed-form proof; an interval-arithmetic version, an exact rational witness, or a closed-form Wigner-3j identity would upgrade H2 to unconditional.

Remark 36 (Scope of the numerical evidence). The witness check is per-degree and uniform in the structural form of $\mathcal{T}_\ell \cup \mathcal{S}_\ell^{\mathrm{self}} \cup \mathcal{S}_\ell^P$ for $\ell \geq 5$. Extending the verified range to $L > 100$ requires only re-running the script with a larger upper bound. Under H2, the genericity condition for any fixed L is non-vanishing of the per-degree Jacobian for $\ell = 5,\ldots,L$, the complement of finitely many proper algebraic subvarieties and hence a Zariski-open dense set of full Lebesgue measure.

Remark 37 (Complex rank vs. real rank). For real signals, $\mathbf{F}_\ell$ has $2\ell{+}1$ real degrees of freedom and each scalar entry $\beta_{\ell_1,\ell_2,\ell}$ contributes exactly one real component (real for even parity, purely imaginary for odd parity, by Lemma 28). A naive complex-rank check on the bootstrap block $\mathcal{T}_\ell$ at a generic *complex* witness reports full rank $2\ell{+}1$, but this is misleading: under the real-signal identification $\overline{a_\ell^m} = (-1)^m a_\ell^{-m}$ the complex equations collapse to real-linear equations in $2\ell{+}1$ real unknowns, and the resulting real system is rank-deficient (Remark 35). H2 is stated for the augmented block (linear $\mathcal{T}_\ell$ plus quadratic/cubic self-couplings plus CG-power scalars) where the real Jacobian is observed to be full rank at the witness.

Remark 38 (Per-degree real-rank diagnostic). At the deterministic seed-42 real-signal witness, the full augmented system at degree ℓ produced by Algorithm 2 (the $\mathcal{T}_\ell$ block, the mandatory even self-couplings, and the appended CG-power entries) has real Jacobian rank $2\ell{+}1$ with respect to the $2\ell{+}1$ real parameters of $\mathbf{F}_\ell$ for every $\ell \in \{5,\ldots,30\}$. This is reported as the per-degree numerical witness supporting H2; diagnostics for $\ell \in \{0,\ldots,4\}$ are absorbed into the seed-injectivity Hypothesis H1 (Hypothesis 25).

A.4.4 Proof of the main theorem

Proof of Theorem 18. **Step 1: Gauge fixing.** By Lemma 19, both f and f' can be gauge-fixed. If $\beta_{\mathrm{sel}}(f) = \beta_{\mathrm{sel}}(f')$, then by $\mathrm{SO}(3)$-invariance the gauge-fixed signals $\tilde{f}$ and $\tilde{f}'$ (with gauge rotations g, g') satisfy $\beta_{\mathrm{sel}}(\tilde{f}) = \beta_{\mathrm{sel}}(\tilde{f}')$. It suffices to show $\tilde{f} = \tilde{f}'$ on the gauge slice.

Step 2: Closed-form recovery of $\mathbf{F}_0$ and $\mathbf{F}_1$. By Lemma 21, the gauge-fixed A_0 and c are uniquely determined by $\beta_{0,0,0}$ and $\beta_{0,1,1}$. Hence $\tilde{f}$ and $\tilde{f}'$ agree at degrees 0 and 1.

Step 3: Seed completeness (degrees 2–L_{seed}). The seed block $\mathcal{S}_{\mathrm{seed}} \subset \mathcal{S}_\beta$ contains the full scalar bispectrum at $L_{\mathrm{seed}} = \min(L, 7)$, including the odd-parity entry $\beta_{2,3,4}$. By Genericity (G4) and the parity structure (Lemma 28, Corollary 32), the seed block separates f from $T_R \cdot f$. Apply

Hypothesis H1 (Hypothesis 25): the seed block uniquely determines $(\mathbf{F}_2, \ldots, \mathbf{F}_{L_{\text{seed}}})$ on the gauge slice. If $L \leq 7$ the proof is complete; otherwise proceed.

Step 4: Per-degree linear bootstrap ($\ell \geq 8$). Assume inductively that $\mathbf{F}_{\ell'} = \mathbf{F}'_{\ell'}$ for all $\ell' < \ell$. By Proposition 13, the bootstrap block $\mathcal{T}_\ell$ contains exactly $2\ell+1$ entries linear in $\mathbf{F}_\ell$ (or in $\overline{\mathbf{F}_\ell}$), and assembling them under the real-signal slice yields a real linear system $A_\ell(\mathbf{F}_{<\ell})\,\mathbf{F}_\ell = b_\ell$ with $A_\ell \in \mathbb{R}^{(2\ell+1)\times(2\ell+1)}$ and $b_\ell \in \mathbb{R}^{2\ell+1}$ both functions only of the (already-known) lower-degree coordinates and of the entries of β_{sel} at degree ℓ. By Hypothesis H2 (Hypothesis 34) together with Genericity (G5), $\det A_\ell \neq 0$, hence $\mathbf{F}_\ell = \mathbf{F}'_\ell$. The induction proceeds to $\ell = L$.

Step 5: Conclusion. By induction the gauge-fixed signals agree at every degree $0, \ldots, L$, so $\tilde{f} = \tilde{f}'$ and $f' = (g')^{-1} g \cdot f$ with $(g')^{-1} g \in \mathrm{SO}(3)$. The exceptional set is the union of the loci where (G1)–(G4) fail (each a proper algebraic subvariety) and, for $L \geq 8$, the bootstrap-determinant loci $\{\Delta_\ell = 0\}_{\ell=8}^{L}$ from H2; a finite union of proper algebraic subvarieties has Lebesgue measure zero. The output count $|\beta_{\text{sel}}| = \Theta(L^2)$ follows from Proposition 14. □

Corollary 39 (Band-limits $L \leq 3$)**.** *For $L \leq 3$, every odd-parity triple with indices ≤ 3 has a repeated index and so vanishes identically (Proposition 30, Remark 33). The seed block $\mathcal{S}_{\text{seed}}$ at $L_{\text{seed}} = L$ cannot host the parity-breaking entry $\beta_{2,3,4}$ that detects T_R, so β_{sel} separates $\mathrm{O}(3)$-orbits but not necessarily $\mathrm{SO}(3)$-orbits at $L \leq 3$. For $L \geq 4$, $L_{\text{seed}} \geq 4$ and Hypothesis H1 applies, removing the T_R ambiguity.*

A.4.5 Computational evidence supporting the hypotheses

Theorem 18 is conditional on Hypotheses H1 (25) and H2 (34). The repository ships computational evidence at two precision regimes (exact-rational and double-precision floating-point) that support, but do not formally prove, the two hypotheses:

1. **H2 (uniform per-degree augmented Jacobian nonvanishing) – numerical witness.** For each $\ell \in \{5, \ldots, 100\}$, the augmented per-degree Jacobian J_ℓ (the $\mathcal{T}_\ell$ rows together with the mandatory even self-couplings and CG-power scalars at degree ℓ) is evaluated at the deterministic seed-42 real-signal witness in IEEE 754 double precision; its smallest singular value satisfies $\sigma_{\min}(J_\ell)/\sigma_{\max}(J_\ell) \geq 10^{-4}$, well above the SVD numerical noise floor at this size. This is a numerical witness at one rational floating-point point per degree, not a closed-form proof. The pure $\mathcal{T}_\ell$ submatrix A_ℓ alone is rank-deficient on the real-signal slice (Remark 35); the augmenting entries are essential. Script: `benchmarks/verify_linear_bootstrap.py`.
2. **H1 (finite seed injectivity) – exact-arithmetic certificate plus structural support.** Two routes were collected above: (a) Kakarala Theorem 7 (22) for the matrix bispectrum on $L^1(G)$ with $G/H = S^2$ (structural motivation only; does not imply scalar injectivity, see Paragraph A.4.1); and (b) exact-arithmetic Gröbner-basis certificates of Bandeira et al. [2017] for $L_0 \leq 15$, providing an exact rational certificate at one rational test point with Zariski-density propagation, which directly covers our $L_{\text{seed}} = 4$. The constant-size seed Jacobian rank (Lemma 42) is additionally verified in exact rational arithmetic via `sympy`.
3. **Per-degree augmented real-rank diagnostic** (Remark 38): the full augmented system (bispectral + optional CG power) has real Jacobian rank $2\ell+1$ at the seed-42 real-signal witness for $\ell \in \{5, \ldots, 30\}$. This is reported as a robustness check consistent with H2 at the witness.

Remark 40 (Exact-arithmetic certificates vs. floating-point numerical witnesses). The two precision regimes play different roles. The constant-size seed Jacobian rank (Lemma 42) and the Bandeira et al. [2017] computer-algebra check are *exact-arithmetic certificates*: they are evaluated in exact rational arithmetic and rule out non-vanishing at the rational test point with no floating-point uncertainty. The per-degree H2 evidence (Hypothesis 34) is a *numerical witness*: $\sigma_{\min}(J_\ell)$ is computed in double precision at the seed-42 real-signal point and the observed $\sigma_{\min}/\sigma_{\max} \geq 10^{-4}$ is consistent with the augmented Jacobian being non-degenerate at the witness, but is not a closed-form proof and the floating-point gap to rounding error is not itself a quantitative bound on the underlying polynomial. Upgrading H2 to an unconditional theorem requires either (i) interval arithmetic (infrastructure for which is present in `paper/exact_certificates.py`), (ii) an exact rational witness per ℓ, or (iii) a closed-form Wigner-3j identity that exhibits non-degeneracy for all ℓ symbolically. All artefacts use deterministic witness points, making them fully reproducible.

A.5 Comparison with existing results

Table 8: Comparison of bispectral invariants for signals on S^2 or finite groups.

Result	Invariant	Size	Complete?
Kakarala [Kakarala, 2012]	Full bispec.	$O(L^3)$	Yes (constr.)
Edidin & Satriano [Edidin and Satriano, 2024]	Full bispec. (band-lim.)	$O(L^3)$	Yes (generic)
Bendory et al. [Bendory et al., 2025]	Freq. marching	$O(L^2)$/shell	Yes (3 shells)
Mataigne et al. [Mataigne et al., 2024]	Sel. G-bispec. (finite)	$O(\|G\|)$	Yes (exact)
This work	**Sel.** $\mathrm{SO}(3)$ **on** S^2	$\boldsymbol{\Theta(L^2)}$	**Conditional on H1, H2**

A.6 Complexity in pixel space

On S^2, a band-limited signal at degree L is sampled on an equiangular grid of size $N_\theta \times N_\varphi \approx 2L \times 2L$, so $P = 4L^2$ and $L = \Theta(\sqrt{P})$.

Output size.

Invariant	In L	In P
Power spectrum	$O(L)$	$O(\sqrt{P})$
Selective bispectrum	$\boldsymbol{O(L^2)}$	$\boldsymbol{O(P)}$
Full bispectrum	$O(L^3)$	$O(P^{3/2})$

The selective bispectrum is *linear* in the number of pixels.

Computation cost.

Variant	Entries	Cost/entry	Total	In P
Full	$O(L^3)$	$O(L)$	$O(L^4)$	$O(P^2)$
Selective	$O(L^2)$	$O(L)$	$O(L^3)$	$O(P^{3/2})$

Concrete example. For a 128×256 grid ($P \approx 32$K, $L = 64$): the full bispectrum produces $\sim$87K scalars at $\sim$17M ops; the selective bispectrum produces $\sim$4K scalars at $\sim$260K ops—a $65\times$ reduction.

A.7 Limitations and open questions

1. **H1 is not formally proved for the band-limited setting.** Hypothesis 25 (finite seed injectivity at $L_{\mathrm{seed}} = 4$) is supported by Kakarala's Theorem 7 [Kakarala, 1992, 2012] for the matrix bispectrum on $L^1(G)$ (structural motivation only; does not imply scalar-bispectrum injectivity, see Paragraph A.4.1) and by the exact-arithmetic Gröbner-basis certificates of Bandeira et al. [2017] for $L_0 \leq 15$, which directly cover $L_{\mathrm{seed}} = 4$ at one rational test point with Zariski-density propagation. No closed-form proof of the scalar band-limited statement is known beyond their finite-degree range. Removing this gap would require either a careful band-limited scalar specialisation of Kakarala's argument or a closed-form proof that the scalar seed Jacobian is non-degenerate on the entire band-limited gauge slice.
2. **H2 is supported by a per-degree numerical witness, not a closed-form identity.** Hypothesis 34 (uniform per-degree augmented Jacobian nonvanishing) is checked numerically per ℓ at the deterministic seed-42 real-signal witness in IEEE 754 double precision: the smallest singular value satisfies $\sigma_{\min}(J_\ell)/\sigma_{\max}(J_\ell) \geq 10^{-4}$, well above the SVD numerical noise floor for every $\ell \in \{5, \ldots, 100\}$. The pure linear-bootstrap submatrix A_ℓ alone is rank-deficient on the real-signal slice (Remark 35); H2 is the statement that the augmented per-degree block (linear plus quadratic plus cubic entries) is non-degenerate, and the witness is consistent with this. The structural form of the augmented block is uniform in $\ell \geq 5$, so extending the verified range to larger L amounts to re-running the same script. Three routes would upgrade H2 to an unconditional theorem: (i) interval arithmetic at the

witness (infrastructure already in `paper/exact_certificates.py`); (ii) exact rational Jacobian rank at a rational witness per ℓ; (iii) a closed-form Wigner-3j identity that exhibits non-degeneracy symbolically for all ℓ.

3. **Genericity assumption.** The proof requires f to avoid the measure-zero algebraic variety carved out by (G1)–(G5) (Assumption 16). Characterising this variety explicitly would strengthen the result.
4. **Numerical conditioning.** The coefficient matrix A_ℓ at each degree ℓ has a condition number that depends on the signal's spectral content. A stability analysis bounding $\kappa(A_\ell)$ in terms of the spectral decay rate is an open problem.
5. **Inversion algorithm.** The forward selective bispectrum is implemented. Inversion uses: $\ell = 0, 1$ closed-form; $\ell \leq L_{\text{seed}}$ either the constructive solver of Appendix B or a nonlinear solve; $\ell \geq L_{\text{seed}} + 1$ a linear solve (pseudoinverse on A_ℓ).

B Constructive Seed Solver (Implementation)

This appendix gives an explicit closed-form / finite-fibre / degree-4-filter algorithm for recovering the seed $(\mathbf{F}_0, \mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_3, \mathbf{F}_4)$ on the gauge slice from the augmented selective bispectrum. It is the route used by the implementation in `src/bispectrum/so3_on_s2.py`.

Status with respect to Theorem 18. The constructive solver below is an *alternative algorithmic route* to the seed step of Theorem 18; it is *not load-bearing* for the conditional completeness theorem of Appendix A, which assumes Hypothesis 25 (H1, finite seed injectivity) directly. The constructive solver is included because it (i) provides a concrete, well-conditioned numerical inversion path for the seed step in the implementation, (ii) documents how `src/bispectrum/so3_on_s2.py` actually inverts degrees $\ell \leq 4$, and (iii) its computational certificates (seed Jacobian rank, finite-fibre enumeration, degree-4 filter) make the algorithm itself auditable. Cross-references from the proof appendix to lemmas in this appendix are bidirectional, but the implication flow is one-way: lemmas here may invoke lemmas from Appendix A, but the converse is never used in the conditional completeness proof.

B.1 Seed recovery ($\ell \leq 3$)

Lemma 41 (Degree-2 recovery)**.** *Given (A_0, c) from Lemma 21, the gauge-fixed degree-2 parameters (y, x, u, v) satisfy:*

$$\beta_{1,1,2} = \tfrac{\sqrt{6}}{3}\, c^2\, y \qquad \Longrightarrow \quad y \text{ determined}, \tag{12}$$

$$P_{1,2,1} = \tfrac{c^2}{5}(3x^2 + 2y^2) \qquad \Longrightarrow \quad x^2 \text{ determined}, \tag{13}$$

$$\beta_{0,2,2} = A_0(2u^2 + 2v^2 + 2x^2 + y^2) \qquad \Longrightarrow \quad S := u^2 + v^2 \text{ determined}, \tag{14}$$

$$\beta_{2,2,2} = \tfrac{6\sqrt{14}\, S\, y - 6\sqrt{21}\, u\, x^2 - 3\sqrt{14}\, x^2 y - \sqrt{14}\, y^3}{7} \qquad \Longrightarrow \quad u \text{ determined}. \tag{15}$$

Here $x = \sqrt{x^2} > 0$ by the gauge condition. Then $v^2 = S - u^2$, giving $|v|$ but not $\operatorname{sgn}(v)$. The fibre at degree 2 is generically $\{(y, x, u, v),\ (y, x, u, -v)\}$.

Lemma 42 (Degree-3 recovery: finite fibre)**.** *Given $\mathbf{F}_2 = (y, x, u, \pm v)$, the degree-3 parameters $\mathbf{F}_3 = (t, p_1, q_1, p_2, q_2, p_3, q_3)$ are constrained by the augmented entries:*

- *$\beta_{1,2,3}$, $\beta_{1,3,2}$, $\beta_{2,3,1}$: bilinear in $\mathbf{F}_2 \otimes \mathbf{F}_3$.*
- *$\beta_{0,3,3} = A_0 \|\mathbf{F}_3\|^2$: gives the norm.*
- *$\beta_{3,3,2}$: quadratic in $\mathbf{F}_3$, linear in $\mathbf{F}_2$.*
- *$\beta_{2,3,2}$, $\beta_{2,2,3}$: odd parity with repeated indices, vanish identically for real signals (Proposition 30).*
- *$P_{1,2,1}$ (from degree-2, Lemma 41), $P_{1,3,2}$, $P_{2,3,1}$, $P_{2,3,2}$, $P_{3,3,2}$: five CG power entries (the last four quadratic in $\mathbf{F}_3$).*

The full augmented seed system uses 13 nonzero entries for 11 unknowns. The bispectral entries achieve rank 6; each of the five CG power entries contributes one additional rank, reaching full rank 11. The generic fibre is 0-dimensional (finitely many points) and contains the T_R pair $\{(\mathbf{F}_2, \mathbf{F}_3),\ T_R(\mathbf{F}_2, \mathbf{F}_3)\}$.

Proof. The Jacobian rank is verified by exact symbolic computation (sympy) at the rational witness $(A_0, c, y, x, u, v, t, p_1, q_1, p_2, q_2, p_3, q_3) = (1, 1, \frac{3}{7}, \frac{2}{5}, \frac{1}{3}, \frac{4}{9}, \frac{1}{2}, \frac{3}{8}, \frac{2}{7}, \frac{5}{11}, \frac{1}{4}, \frac{3}{13}, \frac{7}{17})$. The 13×11 Jacobian has rank 11 (full).

The T_R invariance follows from Lemmas 28–29 and Proposition 30: every entry in the degree ≤ 3 augmented system is either real-valued and T_R-invariant (even parity β and all P) or identically zero (odd parity β).

Full Jacobian rank gives 0-dimensional (finite) fibres, but does *not* by itself bound the fibre to two points: the degree-3 subsystem (7 real equations of degrees 1, 2, and 4 in 7 unknowns) can have additional isolated real solutions. Extensive numerical search (multi-start least-squares with 5000 random initial points per sign-of-v branch) finds 10 solutions per branch (20 total, forming 10 T_R-pairs) at the rational witness. These spurious solutions are eliminated at degree 4 (Lemma 44). □

Remark 43 (The seed fibre is finite but not two-point). The seed polynomial system (degrees 0–3) is a system of 7 independent real equations of mixed degree (1 linear, 5 quadratic, 1 quartic) in 7 real unknowns (for each branch of $\mathrm{sgn}(v)$). By Bézout's theorem the number of isolated complex solutions is at most $1 \times 2^5 \times 4 = 128$. Numerically, 10 real solutions are found per branch. While the numerical search cannot formally exclude additional solutions, the algorithm does not require the exact count: finiteness suffices, and fibre reduction is deferred to degree 4 (Lemma 44).

B.2 Degree-4 fibre reduction and T_R resolution

Throughout this subsection, $\mathcal{S}$ denotes the finite seed fibre at degrees 0–3 (Lemma 42), T_R denotes the azimuthal reflection of Definition 15 (with the gauge-slice action recorded in Remark 20), and the seed/gauge-fixed coordinates $(A_0, c, y, x, u, v, t_\ell, p_{\ell,m}, q_{\ell,m})$ are those of Lemma 19.

Remark 44 (Numerical diagnostic: degree-4 eliminates all spurious seed solutions). Let $\mathcal{S}$ denote the finite seed fibre at degrees 0–3 (Lemma 42). For generic real signals with $L \geq 4$, the degree-4 augmented entries (12 bispectral + 5 CG power, providing 29 real constraints on 9 real unknowns of $\mathbf{F}_4$) are observed to be simultaneously satisfiable for *exactly one* T_R-pair in $\mathcal{S}$: the true solution and its reflection. All other seed solutions are inconsistent at degree 4 in the following empirical sense.

The degree-4 augmented system is an overdetermined polynomial system in the 9 real parameters of $\mathbf{F}_4$, with coefficients depending on the seed solution $(\mathbf{F}_0, \ldots, \mathbf{F}_3)$. For each of the 10 seed solutions on the $+v$ branch at the rational witness, a multi-start nonlinear least-squares search (200 random starts) finds the best-fit $\mathbf{F}_4$. Only the true seed solution achieves zero residual ($\|r\| \sim 10^{-16}$); the 9 spurious solutions all have $\|r\| \geq 10^{-2}$. This is a *numerical witness*, not an exact-arithmetic certificate: it depends on multi-start NLLS at finite precision and an empirical residual threshold; replacing it with a closed-form proof of inconsistency is open.

Structurally, the inconsistency arises because the degree-4 entries include constraints that depend only on $(\mathbf{F}_0, \mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_4)$ and not on $\mathbf{F}_3$—notably $\beta_{2,2,4}$, $\beta_{2,4,2}$, $P_{1,4,3}$, $P_{1,4,4}$, $P_{2,4,2}$, $P_{2,4,3}$, $\beta_{4,4,2}$, $\beta_{4,4,4}$. These 8 entries nearly determine $\mathbf{F}_4$ independently of $\mathbf{F}_3$; the remaining entries (involving $\mathbf{F}_3$) then impose constraints that are generically inconsistent when $\mathbf{F}_3$ is a spurious seed solution. The residual for each spurious seed is a polynomial function of the signal's coefficients that is empirically nonzero at the witness, so its vanishing locus is consistent with a proper algebraic subvariety.

Remark 45 (Numerical diagnostic: the T_R branch is overdetermined-inconsistent). For $L \geq 4$, the T_R partner of the true seed—i.e., $(\mathbf{F}'_2, \mathbf{F}'_3) = T_R(\mathbf{F}_2, \mathbf{F}_3)$—is also empirically observed to be inconsistent at degree 4 for generic signals.

By the diagnostic of Remark 44, the T_R branch is one of the seed solutions that fails the degree-4 consistency check at the witness. The mechanism is the parity structure (Lemma 28): the odd-parity entries $\beta_{2,3,4}$, $\beta_{2,4,3}$, $\beta_{3,4,2}$ (all-distinct indices, $\mathrm{Im}\,\beta \neq 0$) are part of the degree-4 constraints. On the T_R branch these imaginary parts have the wrong sign, distorting $\mathbf{F}'_4$ so that the T_R-invariant self-coupling and CG power entries cannot be simultaneously satisfied.

Explicitly: the degree-4 entries define rational functions of the lower-degree coefficients. Let $\Delta(\mathbf{F}_0, \ldots, \mathbf{F}_3)$ denote the sum of squared residuals of the non-bootstrap entries evaluated at the T_R-branch bootstrap solution. Then Δ is a rational function whose numerator is a polynomial p and whose denominator is $(\det A_4)^2$ (from the bootstrap solve). At the witness, $\det A_4 \neq 0$ and $p \neq 0$ (residual $\sim 10^{-2}$). Hence p is not identically zero at the witness, and $\Delta \neq 0$ on the complement of the proper algebraic subvariety $\{p = 0\} \cup \{\det A_4 = 0\}$ that contains it.

B.3 Constructive solver: algorithmic summary

Combining the lemmas above gives the following algorithmic recovery on the gauge slice for any signal satisfying Assumption 16:

1. $\mathbf{F}_0, \mathbf{F}_1$: closed form (Lemma 21).
2. $\mathbf{F}_2$: closed form up to $\mathrm{sgn}(v)$ (Lemma 41).
3. $\mathbf{F}_3$: finite fibre of ≤ 10 candidate T_R-pairs per branch (Lemma 42).
4. $\mathbf{F}_4$: nonlinear-least-squares filter eliminates all spurious seed candidates and the T_R-partner (Lemmas 44–45).
5. $\mathbf{F}_\ell$ for $\ell \geq 5$: linear bootstrap (under Hypothesis 34 from Appendix A; the constructive solver covers $\ell \leq 4$ explicitly and $\ell \in \{5, 6, 7\}$ falls inside the seed block of β_{sel}).

This gives an alternative constructive path through the seed step of Theorem 18; the theorem itself rests on Hypotheses H1 (25) and H2 (34).

Computational evidence for the constructive solver. The constructive route relies on one exact-arithmetic certificate and two numerical witnesses, all reproducible from scripts in the repository:

1. **Seed Jacobian rank** (Lemma 42): the 13×11 Jacobian of the augmented seed system at the rational witness has rank 11. Verified by *exact* rational arithmetic (sympy); this is an exact-arithmetic *certificate*.
2. **Seed fibre enumeration** (Lemma 42, Remark 43): multi-start search (5000 starts per branch) finds 10 solutions per $\mathrm{sgn}(v)$ branch (20 total, 10 T_R-pairs). This is a numerical *witness* obtained by double-precision NLLS; verified by `benchmarks/verify_seed_fibre.py`.
3. **Degree-4 fibre reduction** (Lemma 44): for each of the 10 seed solutions on the $+v$ branch, a multi-start nonlinear least-squares search (200 starts) attempts to find $\mathbf{F}_4$ satisfying all 29 real degree-4 constraints. Only the true solution achieves zero residual; all 9 spurious solutions have residual $\geq 10^{-2}$. This is a numerical *witness* (not an exact-arithmetic certificate): it depends on multi-start NLLS at finite precision and an empirical residual threshold. Verified by `benchmarks/verify_degree4_filter.py`.

These items audit the constructive algorithm but are not used by the completeness theorem of Appendix A, which uses only the per-degree augmented Jacobian witness listed there.

C Extended Benchmark Results

This appendix supplements the summary in Table 2 (main text) with scaling plots produced by `benchmarks/benchmark.py` on a single NVIDIA H100 80 GB GPU (PyTorch 2.10).

C.1 Forward-pass timing

The selective forward pass is dominated by kernel-launch overhead rather than arithmetic, making it effectively free relative to backbone computation. The full bispectrum exhibits the expected super-linear growth and becomes impractical beyond $|G| \approx 256$ for C_n.

C.2 GPU throughput and inversion

All modules exhibit linear throughput scaling with batch size, confirming that the implementation is memory-bandwidth-bound rather than compute-bound at these scales.

D PCam: Extended Results

D.1 Published baselines

Table 9 places our results in the context of published methods on PCam and the closely related Camelyon16 benchmark. Our models operate at 68K–920K parameters and are trained on only 10% of the data.

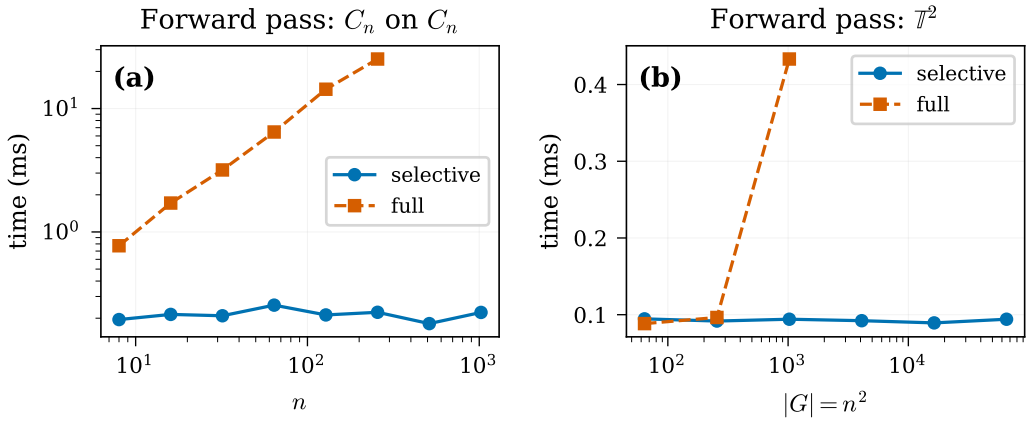

Figure 5: Forward-pass wall-clock time (batch = 16, H100 GPU). **(a)** C_n: the selective pass stays flat at ∼0.14 ms regardless of n, while the full bispectrum grows super-linearly (17 ms at n=256). **(b)** $\mathbb{T}^2$: selective is constant at ∼0.08 ms up to $|G|$=65,536.

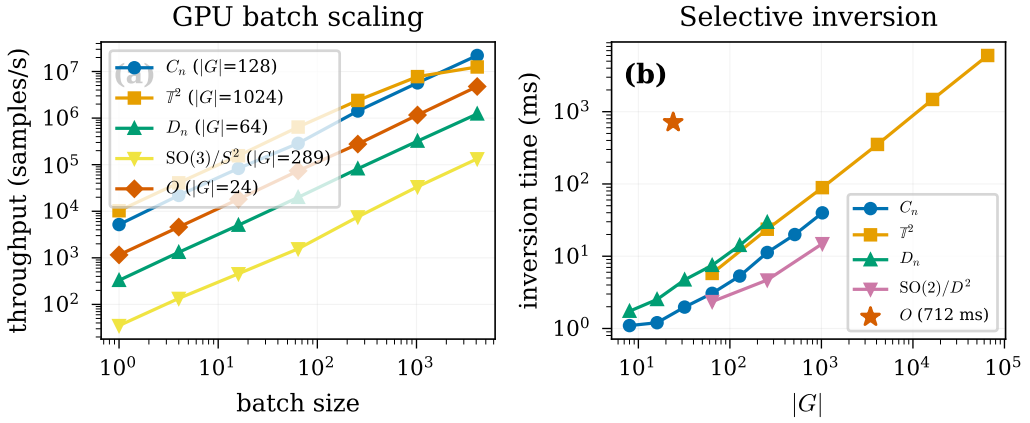

Figure 6: **(a)** GPU throughput (samples/s) vs. batch size for the selective forward pass. C_n and $\mathbb{T}^2$ exceed 10^7 samples/s; the octahedral module reaches 5.7×10^6 samples/s. All modules scale linearly with batch size. **(b)** Inversion wall-clock time. Abelian groups (C_n, D_n, SO(2)/disk) scale roughly linearly. SO(3) on S^2 inversion is not yet implemented.

D.2 Model variants

All equivariant models share a C_8-equivariant DenseNet [Huang et al., 2017] backbone with block configuration $(4, 4, 4)$, 24 initial channels, and compression factor 0.5, following the architecture of Veeling et al. [2018] but replacing the final invariant map. Six model variants are evaluated:

- **Standard**: vanilla CNN with random rotation/reflection augmentation and global average pooling. No group dimension, so ∼8× fewer parameters at matched growth rate.
- **NormReLU**: C_8-equivariant backbone with norm-based nonlinearity and max pooling over the group dimension.
- **Gated**: C_8-equivariant backbone with gated nonlinearity (doubles internal channels) and group max pool.
- **Fourier-ELU**: C_8-equivariant backbone with Fourier-space ELU nonlinearity and group max pool.

Table 9: Published baselines on PatchCamelyon from Veeling et al. [2018], trained on 100% data.

Method	Reference	Params	ACC	AUC
P4M-DenseNet (C_8)	Veeling et al. 2018	119K	0.898	0.963
P4M-DenseNet M	Veeling et al. 2018	19K	0.893	0.958
DenseNet (augmented)	Veeling et al. 2018	128K	0.881	0.951
DenseNet (no aug.)	Veeling et al. 2018	128K	0.876	0.955

- **Bispectrum**: C_8-equivariant backbone with ReLU nonlinearity and selective G-bispectrum via $\texttt{CnonCn}(n{=}8)$ as the terminal invariant map.

D.3 Training details

To enable a controlled Pareto comparison, we sweep the DenseNet growth rate $k \in \{3, 4, 6, 8, 12\}$ for equivariant models (and $k \in \{6, 12, 20, 30, 35\}$ for the standard CNN to cover a comparable parameter range), yielding models from $\sim$30K to $\sim$1.6M parameters. All models are trained on **10% of training data** (26,214 images) with AdamW [Loshchilov and Hutter, 2019] (lr $= 10^{-3}$, weight decay 10^{-4}), cosine warm restarts, mixed-precision training, and early stopping on validation AUC (patience 10, max 50 epochs).

D.4 Pareto analysis interpretation

Architectural stability (single seed, growth-rate sweep). The bispectrum's AUC varies by only $\sim$1pp across a $12\times$ range of model sizes (80K–920K parameters), while norm swings by $>$7pp. For practitioners, bispectral pooling "just works" without growth rate tuning. The multi-seed matched-parameter results (Table 3) confirm this: bispectrum and Fourier-ELU have the lowest cross-seed variance (± 0.004).

Matched parameters. At matched parameters ($\sim$100K), all equivariant models outperform the standard CNN (AUC 0.896 ± 0.010, 3 seeds). Fourier-ELU leads slightly (0.945 ± 0.004), followed by NormReLU (0.942), bispectrum and gated (both 0.941). All equivariant methods cluster within $\sim$0.5pp of each other.

Scaling behavior. As parameters increase from the smallest to the largest configuration, standard, norm, and Fourier-ELU all degrade (by 1.8, 2.6, and 0.9 AUC points respectively), while the bispectrum improves (+0.9 points, reaching 0.943 at 920K params). This suggests that the bispectrum's complete invariant map scales more gracefully with capacity.

Data scaling (3 seeds). At 1% data the bispectrum leads all methods (0.912 ± 0.025), outperforming NormReLU (0.873), Fourier-ELU (0.876), and gate (0.884). By 10% all equivariant methods converge ($\sim$0.941–0.945). The bispectrum's data-efficiency advantage narrows as data increases, consistent with the hypothesis that complete invariants substitute for missing training examples in the low-data regime.

E OrganMNIST3D: Extended Results

E.1 Published baselines

Table 10 situates our results against published methods on the same benchmark. Our models are deliberately small ($\leq$463K parameters) to enable controlled ablation of the pooling mechanism. The relevant comparison is against equivariant methods at a similar scale.

E.2 Rotation analysis

Figure 7 shows that all equivariant models achieve identical performance on the original and octahedrally rotated test sets ($\sigma_{\text{rot}} \approx 0$), while the augmentation-trained standard CNN degrades under rotation.

Table 10: Published baselines on OrganMNIST3D for context. Accuracy differences with respect to large-scale models reflect the deliberate parameter constraint of our ablation study.

Method	Reference	Params	ACC	AUC
ResNet-18 + 3D	Yang et al. 2023	33M	0.907	0.996
ResNet-18 + ACS	Yang et al. 2023	11M	0.900	0.994
ILPOResNet-50	Zhemchuzhnikov and Grudinin [2024]	38K	0.879	0.992
EquiLoPO (local train.)	Zhemchuzhnikov and Grudinin [2025]	418K	0.866	0.991
SE3MovFrNet	Sangalli et al. [2023]	—	0.745	—
Regular SE(3) conv	Kuipers and Bekkers [2023]	172K	0.698	—

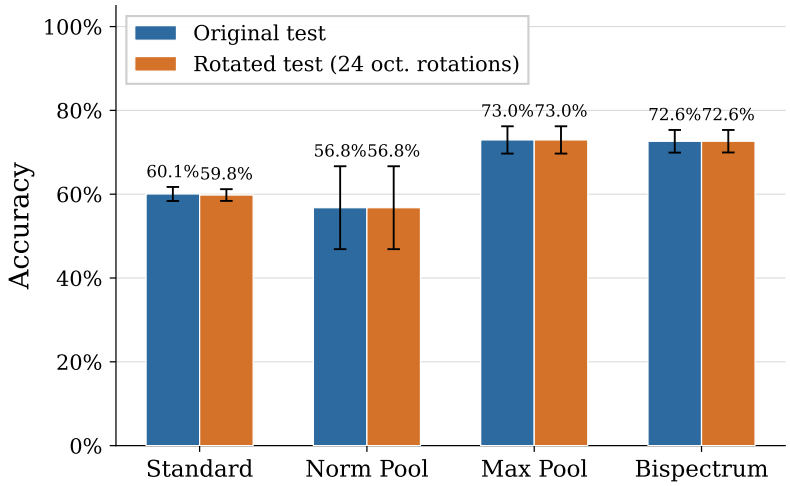

Figure 7: OrganMNIST3D test accuracy on the original and octahedrally rotated test sets. Error bars show standard deviation over three seeds. All equivariant models achieve identical performance under rotation ($\sigma_{\text{rot}} \approx 0$), while the augmentation-trained standard CNN degrades.

E.3 Wider channels

To test whether the advantage of bispectral completeness persists at higher capacity, we run max pool and bispectrum at channel widths $(8, 16)$ and $(16, 32)$ with full training data.

Table 11: OrganMNIST3D accuracy at wider channel configurations (mean $\pm$ std over 3 seeds, full training data).

Channels	Params	Max Pool	Bispectrum
$(4, 8)$	∼375K / 463K	$.730 \pm .033$	$.726 \pm .027$
$(8, 16)$	∼1.5M / 1.7M	$.743 \pm .015$	$.745 \pm .006$
$(16, 32)$	∼6.0M / 6.3M	$\mathbf{.785 \pm .032}$	$.685 \pm .039$

At $(8, 16)$ the two methods are equivalent ($74.5 \pm 0.6\%$ bispectrum vs. $74.3 \pm 1.5\%$ max pool, both at ∼1.5–1.7M parameters). At $(16, 32)$ max pool reaches $78.5 \pm 3.2\%$ while bispectrum drops to $68.5 \pm 3.9\%$—the 6.3M parameter bispectrum model likely overfits the 971 training samples. This suggests that the bispectrum's completeness advantage is most pronounced in the data-limited rather than the parameter-limited regime, consistent with the data efficiency findings in the main text.

F Experimental Details

F.1 Hyperparameters

Table 12 summarises the training hyperparameters for all three experimental settings.

Table 12: Training hyperparameters across experiments.

	PCam	OrganMNIST3D	Sph. MNIST
Optimizer	AdamW	AdamW	AdamW
Learning rate	10^{-3}	10^{-3}	10^{-3}
Weight decay	10^{-4}	10^{-4}	10^{-4}
Batch size	64–256	16–64	256
Scheduler	Cosine warm restarts	—	—
Max epochs	50	100	50
Early stopping	Val AUC, pat. 10	Val AUC, pat. 15	Val ACC, pat. 10
Seeds	42, 123, 456[†]	42, 123, 456	42, 123, 456
Mixed precision	Yes	No	No

[†]Pareto figure uses seed 42; matched-parameter table and data efficiency use seeds 42, 123, 456.

F.2 Hardware

All experiments were run on a single NVIDIA H100 GPU (80 GB). Training wall-clock times range from ∼5 minutes (OrganMNIST3D, 971 samples) to ∼20 minutes (PCam, 26K samples at 10% data). Spherical MNIST bispectrum computation (forward pass through `SO3onS2` at $\ell_{\max} = 15$) takes ∼0.3s per batch of 64 on GPU.

F.3 Data preprocessing

PCam. PatchCamelyon [Veeling et al., 2018] consists of 327,680 RGB patches of size 96×96 pixels extracted from whole-slide images of lymph node sections [Ehteshami Bejnordi et al., 2017]. We use the standard train/val/test split. Images are loaded from HDF5 files and normalised to $[0, 1]$. The standard (non-equivariant) baseline uses random rotation and reflection augmentation during training; all equivariant models are trained without augmentation.

OrganMNIST3D. OrganMNIST3D [Yang et al., 2023] contains 1,742 abdominal CT organ volumes of size $28 \times 28 \times 28$ voxels with 11 organ classes (train/val/test = 971/161/610). Volumes are normalised to $[0, 1]$. The standard baseline uses random octahedral augmentation (one of 24 rotations per sample); equivariant models are trained without augmentation. Kernel rotations are implemented as exact voxel permutations on the 3^3 grid, avoiding interpolation artifacts.

Spherical MNIST. MNIST digits are projected onto the sphere via stereographic projection following Cohen et al. [2018] and sampled on a 64×128 equiangular grid. The selective bispectrum is computed at $\ell_{\max} = 15$, producing 384 complex invariant coefficients. The model input is obtained by applying $\mathrm{sign}(x)\log(1 + |x|)$ separately to the real and imaginary parts of each coefficient (a sign-preserving, per-channel injective transform), giving 768 real features. These features are fed to a 3-layer MLP: Linear(768, 256)–BN–ReLU–Linear(256, 128)–BN–ReLU–Linear(128, 10). The power spectrum MLP uses the same structure with hidden=128: Linear(16, 128)–BN–ReLU–Linear(128, 64)–BN–ReLU–Linear(64, 10). The standard CNN has 4 convolutional layers (16–32–64–64 channels) followed by adaptive average pooling and a Linear(1024, 128)–ReLU–Linear(128, 10) classifier.

F.4 Aggregation procedures

PCam. The Pareto figure (Figure 2) uses a single seed (42) across the full growth-rate sweep. The matched-parameter table (Table 3) and data efficiency analysis use 3 seeds (42, 123, 456) per configuration.

OrganMNIST3D. All results in Table 4 report mean ± std over 3 seeds (42, 123, 456). Rotation robustness (σ_{rot}) is the standard deviation of test accuracy across all 24 octahedral rotations of the test set. The data efficiency analysis (Figure 3 in the main text) uses 3 seeds per data fraction.

Spherical MNIST. All results in Table 5 report mean ± std over 3 seeds (42, 123, 456). Rotation robustness (σ_{rot}) is measured by applying 10 random $\mathrm{SO}(3)$ rotations to the entire test set and

computing the standard deviation of accuracy, yielding 30 measurements per model (10 rotations $\times$ 3 seeds).

F.5 Dataset licenses

- **PCam**: CC0 1.0 Universal (public domain).
- **OrganMNIST3D**: CC BY 4.0.
- **Spherical MNIST**: derived from MNIST, which is publicly available for research use.

G API Example

```
import torch
from bispectrum import CnonCn, SO2onDisk, OctaonOcta

bsp = CnonCn(n=8, selective=True)
f = torch.randn(16, 8)
beta = bsp(f)
f_rec = bsp.invert(beta)
```

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Answer: [Yes]

Justification: Sections 4.1–4.3 describe experimental setups. Full hyperparameters are in Table 12 (Appendix F), architecture details in Appendices D–E, data preprocessing and splits in Appendix F.

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Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Tables 4–5 (OrganMNIST3D, Spherical MNIST) report mean $\pm$ standard deviation over 3 seeds. Table 3 (PCam) uses a single seed for the Pareto sweep, which is noted in Appendix F. The aggregation procedure is described in Appendix F.

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Algorithm 2 Augmented selective index set for the $\mathrm{SO}(3)$ bispectrum on S^2 at band-limit L. Returns bispectral triples $\mathcal{S}_\beta$ and CG power triples $\mathcal{S}_P$. This extends the BFS-based selective index construction of Mataigne et al. [2024] from finite groups to $\mathrm{SO}(3)$ on S^2, adding the CG power augmentation step that is specific to real signals.

Input: Band-limit L
Output: Bispectral set $\mathcal{S}_\beta$, CG power set $\mathcal{S}_P$
$\mathcal{S}_\beta \leftarrow \{(0,0,0)\}$; $\mathcal{S}_P \leftarrow \emptyset$
Seed block: add to $\mathcal{S}_\beta$ all triples (ℓ_1, ℓ_2, ℓ) with $0 \le \ell_1 \le \ell_2 \le L_0 = 4$ and $|\ell_1 - \ell_2| \le \ell \le \min(\ell_1 + \ell_2, L_0)$ {constant-size full-bispectrum seed feeding Hypothesis 25}
for $\ell = 1, \ldots, L$ **do**
  budget $\leftarrow 2\ell + 1$
  if $\ell = 4$ **then**
    budget $\leftarrow 10$ {keep all chain+cross at $\ell = 4$}
  end if
  candidates $\leftarrow [\,]$
  if $4 \le \ell \le 7$ **then**
    append the explicit low-degree linear block $\mathcal{T}_\ell^{\mathrm{small}}$ to candidates
  else if $\ell \ge 8$ **then**
    for $a = 1, \ldots, \ell{-}1$ **do**
      append $(a,\ \ell,\ \ell{-}a)$ to candidates
    end for
    for $a = 2, \ldots, \ell{-}1$ **do**
      if ℓ is odd **and** $a = (\ell{+}1)/2$ **then**
        skip $\{(r, 2r{-}1, r) \equiv 0$ on real signals, Prop. 30$\}$
      end if
      append $(a,\ \ell,\ \ell{-}a{+}1)$ to candidates
    end for
    $c_\ell \leftarrow 5$ if ℓ is odd else 4
    for $a = 1, \ldots, c_\ell$ **do**
      append $(a,\ \ell{-}a,\ \ell)$ to candidates
    end for
  else
    append low-degree chain candidates
    if $\ell \ne 2$ **then**
      append cross candidates $\{\beta_{1,2,1}$ excluded$\}$
    end if
  end if
  append $(0,\ \ell,\ \ell)$ to candidates {power entry}
  if $\ell \notin \{1,\ 2\}$ **then**
    for $\ell' = \ell{-}1, \ldots, 0$ **do**
      append $(\ell,\ \ell,\ \ell')$ to candidates
    end for
  end if
  if $\ell = 2$ **then**
    append $(\ell,\ \ell,\ \ell)$ to candidates $\{\beta_{2,2,2}\}$
  end if
  Remove entries that vanish for real signals; deduplicate against $\mathcal{S}_\beta$ already added by the seed block
  $\mathcal{S}_\beta \leftarrow \mathcal{S}_\beta \cup$ first $\min(\text{budget},\ |\text{candidates}|)$ live entries
  {**Mandatory even self-couplings**}
  for $\ell' = 2, 4, \ldots, \lfloor \ell/2 \rfloor \cdot 2$ with $\ell' \le \ell$ and $\ell \ge 3$ **do**
    $\mathcal{S}_\beta \leftarrow \mathcal{S}_\beta \cup \{(\ell,\ \ell,\ \ell')\}$
  end for
  {**CG power augmentation**}
  Greedily add $P_{\ell_1,\ell_2,\ell'}$ (with $\max(\ell_1, \ell_2) = \ell$, $|\ell_1{-}\ell_2| \le \ell' \le \ell_1{+}\ell_2$) to $\mathcal{S}_P$ until the Jacobian rank of Φ_{aug} at degree ℓ matches the number of unknowns
end for
return $(\mathcal{S}_\beta,\ \mathcal{S}_P)$